Unit 3

Module 3 :

Ordinary Differential Equations of Higher Orders

3

Higher Order Linear Ordinary Differential Equations

Syllabus

Higher order linear differential equations with constant and variable coefficients, Euler-Cauchy equations, solution by variation of parameters, method of undetermined coefficients.

Contents

3.1 Introduction

A differential equation of the form $F\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \ldots, \frac{d^ny}{dx^n}\right) = 0$ in which the dependent variable $y(x)$ and its derivatives viz. $\frac{dy}{dx}, \frac{d^2y}{dx^2}$ etc. occur in first degree and are not multiplied together is called a **linear differential equation.**

3.2 Linear Differential Equations (LDE) with Constant Coefficients

A general linear differential equation of n^{th} order with constant coefficients is given by :

$$k_0 \frac{d^ny}{dx^n} + k_1 \frac{d^{n-1}y}{dx^{n-1}} + \ldots + k_{n-1}\frac{dy}{dx} + k_n y = F(x)$$

where ks are constant and F(x) is a function of x alone or constant.

$$\Rightarrow \quad \left(k_0 D^n + k_1 D^{n-1} + \ldots + k_{n-1}D + k_n\right)y = F(x)$$

Or $f(D)y = F(x)$, where $D^n \equiv \frac{d^n}{dx^n}$, $D^{n-1} \equiv \frac{d^{n-1}}{dx^{n-1}}$, ..., $D \equiv \frac{d}{dx}$ are called differential operators.

3.3 Solving Linear Differential Equations with Constant Coefficients

GTU : Summer-18,20,23,24, Winter-18,19,22,23

Complete solution of equation $f(D)y = F(x)$ is given by y = C.F. + P.I. where C.F. denotes complimentary function and P.I. is particular integral.

when $F(x) = 0$, then solution of equation $f(D)y = 0$ is given by y = C.F.

3.3.1 Rules for Finding Complimentary Function (C.F.)

Consider the equation $f(D)\, y = F(x)$

$$\Rightarrow \quad (k_0 D^n + k_1 D^{n-1} + \ldots k_{n-1}D + k_n)\, y = F(x)$$

Step 1 : Put D = m, auxiliary equation (A.E.) is given by $f(m) = 0$

$$\Rightarrow \quad k_0 m^n + k_1 m^{n-1} + \ldots + k_{n-1}m + k_n = 0 \qquad \ldots(3.3.1)$$

Step 2 : Solve the auxiliary equation given by (3.3.1)

I. If the n roots of A.E. are real and distinct say $m_1, m_2, \ldots m_n$

$$\text{C.E.} = C_1 e^{m_1 x} + C_2 e^{m_2 x} + \ldots + C_n e^{m_n x}$$

II. If two or more roots are equal i.e. $m_1 = m_2 = \ldots = m_k$, $k \le n$

$$\text{C.F.} = \left(C_1 + C_2 x + C_3 x^2 + \ldots + C_k x^{k-1}\right) e^{m_1 x} + \ldots + C_n e^{m_n x}$$

III. If A.E. has a pair of imaginary roots i.e.

$$m_1 = \alpha + i\beta, m_2 = \alpha - i\beta$$

$$\text{C.F.} = e^{ax}(C_1 \cos\beta x + C_2 \sin\beta x) + C_3 e^{m_3 x} + \ldots + C_n e^{m_n x}$$

IV. If 2 pairs of imaginary roots are equal i.e.

$$m_1 = m_2 = \alpha + i\beta$$

$$m_3 = m_4 = \alpha - i\beta$$

$$\text{C.F.} = e^{\alpha x}\left[(C_1 + C_2 x)\cos\beta x (C_3 + C_4 x)\sin\beta x\right] + \ldots + C_n e^{m_n x}$$

Ex. 3.3.1 : *Solve the differential equation :*
$\frac{d^2y}{dx^2}-8\frac{dy}{dx}+15y=0.$

Sol. : $\Rightarrow \left(D^2-8D+15\right)y=0$

Auxiliary equations is : $m^2-8m+15=0$

$\Rightarrow (m-3)(m-5)=0$

$$\Rightarrow \quad m = 3,5$$

$$\text{C.F.} = C_1e^{3x}+C_2e^{5x}$$

Since F(x) = 0, solution is given by y = C.F.

$$\Rightarrow \quad y = \mathbf{C_1e^{3x}+C_2e^{5x}}$$

Ex. 3.3.2 : *Solve the differential equation :*
$\frac{d^3y}{dx^3}-6\frac{d^2y}{dx^2}+11\frac{dy}{dx}-6y=0.$

Sol. : $\Rightarrow (D^3-6D^2+11D-6)y=0$

Auxiliary equations is :

$m^3-6m^2+11m-6=0$ ... (1)

By hit and trial (m – 2) is a factor of (1)

$\therefore$ (1) may be rewritten as

$$m^3-2m^2-4m^2+8m+3m-6 = 0$$

$$\Rightarrow \quad m^2(m-2)-4m(m-2)+3(m-2) = 0$$

$$\Rightarrow \quad (m^2-4m+3)(m-2) = 0$$

$$\Rightarrow \quad (m-3)(m-1)(m-2) = 0$$

$$\Rightarrow \quad m = 1,2,3$$

$$\text{C.F.} = C_1e^{x}+C_2e^{2x}+C_3e^{3x}$$

Since F(x) = 0, solution is given by y = C.F.

$$\Rightarrow \quad y = \mathbf{C_1e^{x}+C_2e^{2x}+C_3e^{3x}}$$

Ex. 3.3.3 : *Solve* $(D^4-10D^3+35D^2-50D+24)\,y=0.$

Sol. : Auxiliary equation is :

$m^4-10m^3+35m^2-50m+24=0$ ..(1)

By hit and trial (m – 1) is a factor of (1)

$\therefore$(1) may be rewritten as

$m^4-m^3-9m^3+9m^2+26m^2-26m-24m+24=0$

$\Rightarrow m^3(m-1)-9m^2(m-1)+26m(m-1)-24(m-1)=0$

$\Rightarrow (m-1)(m^3-9m^2+26m-24)=0$ ...(2)

By hit and trial (m – 2) is a factor of (2)

$\therefore$ (2) may be rewritten as

$(m-1)(m^3-2m^2-7m^2+14m+12m-24)=0$

$\Rightarrow (m-1)[m^2(m-2)-7m(m-2)+12(m-2)]=0$

$\Rightarrow (m-1)(m^2-7m+12)(m-2)=0$

$\Rightarrow (m-1)(m-3)(m-4)(m-2)=0$

$$\Rightarrow \quad m = 1,2,3,4$$

$$\text{C.F.} = C_1e^{x}+C_2e^{2x}+C_3e^{3x}+C_4e^{4x}$$

Since F(x) = 0, solution is given by y = C.F.

$$\Rightarrow \quad y = \mathbf{C_1e^{x}+C_2e^{2x}+C_3e^{3x}+C_4e^{4x}}$$

Ex. 3.3.4 : *Solve the differential equation :*
$\frac{d^3y}{dx^3}+2\frac{d^2y}{dx^2}+\frac{dy}{dx}=0.$

Sol. : $\Rightarrow (D^3+2D^2+D)y=0$

Auxiliary equation is : $m^3+2m^2+m=0$

$$\Rightarrow \quad m(m^2+2m+1) = 0$$

$$\Rightarrow \quad m(m+1)^2 = 0$$

$$\Rightarrow \quad m = 0,-1,-1$$

$$\text{C.F.} = C_1+(C_2+C_3x)\,e^{-x}$$

Since F(x) = 0, solution is given by y = C.F.

$$\Rightarrow \quad y = \mathbf{C_1+(C_2+C_3x)\,e^{-x}}$$

Ex. 3.3.5 : *Solve the differential equation :*
$\frac{d^4y}{dx^4}-2\frac{d^2y}{dx^2}+y=0.$

Sol. : $\Rightarrow (D^4-2D^2+1)\,y=0$

Auxiliary equation is : $m^4-2m^2+1=0$

$$\Rightarrow \quad m(m^2-1)^2 = 0$$

$$\Rightarrow \quad (m+1)^2(m-1)^2 = 0$$

$$\Rightarrow \quad m = -1,-1,1,1$$

$$\text{C.F.} = (C_1+C_2x)\,e^{-x}+(C_3+C_4x)\,e^{x}$$

Since $F(x) = 0$, solution is given by $y = C.F.$

$$\Rightarrow \quad y = (C_1 + C_2 x)e^{-x} + (C_3 + C_4 x)e^{x}$$

Ex. 3.3.6 : *Solve the differential equation :* $\frac{d^3y}{dx^3} - 2\frac{dy}{dx} + 4y = 0.$

Sol. : $\Rightarrow (D^3 - 2D^2 + 4)\, y = 0$

Auxiliary equation is : $m^3 - 2m + 4 = 0$ ...(1)

By hit and trial $(m + 2)$ is a factor of (1)

$\therefore$ (1) may be rewritten as

$$m^3 + 2m^3 - 2m^2 - 4m + 2m + 4 = 0$$

$$\Rightarrow m^2(m+2) - 2m(m+2) + 2(m+2) = 0$$

$$\Rightarrow \quad (m+2)(m^2 - 2m + 2) = 0$$

$$\Rightarrow \quad m = -2, 1 \pm i$$

$$C.F. = C_1 e^{-2x} + e^{x}(C_2 \cos x + C_3 \sin x)$$

Since $F(x) = 0$, solution is given by $y = C.F.$

$$\Rightarrow \quad y = C_1 e^{-2x} + e^{x}(C_2 \cos x + C_3 \sin x)$$

Ex. 3.3.7 : *Solve the differential equation :* $\left(D^2 - 2D + 5\right)^2 y = 0.$

Sol. :

Auxiliary equation is : $\left(m^2 - 2m + 5\right)^2$ ...(1)

Solving (1), we get

$$\Rightarrow \quad m = 1 \pm 2i, 1 \pm 2i$$

$$C.F. = e^{x}\left[(C_1 + C_2 x)\cos 2x + (C_3 + C_4 x)\sin 2x\right]$$

Since $F(x) = 0$, solution is given by $y = C.F.$

$$\Rightarrow \quad y = e^{x}\left[(C_1 + C_2 x)\cos 2x + (C_3 + C_4 x)\sin 2x\right]$$

Ex. 3.3.8 : *Solve the differential equation :* $\left(D^2 + 4\right)^3 y = 0.$

Sol. : Auxiliary equation is : $\left(m^2 + 4\right)^3$ ...(1)

Solving (1), we get

$$\Rightarrow \quad m = \pm 2i, \pm 2i, \pm 2i$$

$$C.F. = \left(C_1 + C_2 x + C_3 x^2\right)\cos 2x + \left(C_4 + C_5 x + C_6 x^2\right)\sin 2x$$

Since $F(x) = 0$, solution is given by $y = C.F.$

$$\Rightarrow \quad y = \left(C_1 + C_2 x + C_3 x^2\right)\cos 2x + \left(C_4 + C_5 x + C_6 x^2\right)\sin 2x$$

Ex. 3.3.9 : *Solve* $y'' - 4y' + 4y = 0$

GTU : Winter-22,19, AEM, Marks 3

Sol. :

Here : $y'' - 4y' + 4y = 0$

$\therefore$ A.E. $D^2 - 4D + 4 = 0$

$$(D-2)^2 = 0$$

$\therefore$ $D = 2, 2$

$\therefore$ $y = (C_1 + C_2 x)\, e^{2x}$

Ex. 3.3.10 : *Solve* $y''' - 6y'' + 11y' - 6y = 0$

GTU : Summer-18, AEM, Marks 3

Sol. : Here, $y''' - 6y'' + 11y' - 6y = 0$ is given

A.E. $D^3 - 6D^2 + 11D - 6 = 0$

$\therefore$ $D^3 - D^2 - 5D^2 + 5D + 6D - 6 = 0$

$\therefore D^2(D-1) - 5D(D-1) + 6(D-1) = 0$

$\therefore$ $(D-1))(D^2 - 5D + 6) = 0$

$\therefore$ $(D-1)(D-2)(D-3) = 0$

$\therefore$ $D = 1, 2, 3$

$\therefore$ $y = C_1 e^x + C_2 e^{2x} + C_3 e^{3x}$

Ex. 3.3.11 : *Solve* $\dfrac{d^3y}{dx^3} - 3\dfrac{dy}{dx} + 2y = 0.$

GTU : Summer-20, AEM, Marks 3

Sol. : Here $y''' - 3y' + 2y = 0$ is given

A.E. $D^3 - 3D^2 + 2 = 0$

$\therefore$ $D^3 - D^2 + D^2 - D - 2D + 2 = 0$

$\therefore D^2(D-1) + D(D-1) - 2(D-1) = 0$

$\therefore$ $(D-1)(D^2 + D - 2) = 0$

$\therefore$ $(D-1)(D^2 + 2D - D - 2) = 0$

$\therefore$ $(D-1)(D(D+2) - 1(D+2)) = 0$

$\therefore$ $(D-1)(D-1)(D+2) = 0$

$\therefore$ $D = 1, 1, -2$

$\therefore$ $y = e^x + (C_1 + C_2 x) + C_3 e^{-2x}$

Ex. 3.3.12 : *Solve* $y'' + 3y' + 2y = 0$

GTU : Summer-24, AEM, Marks 3

Sol. :

Here $y'' + 3y' + 2y = 0$ is given

$\therefore$ $D^2 + 3D + 2 = 0$

$D^2 + 2D + D + 2 = 0$

$\therefore$ $D(D+2) + 1(D+2) = 0$

$\therefore$ $D + 1 = 0,\ D + 2 = 0$

$\therefore$ $D = -1$ and $D - 2$

$\therefore$ $y = C_1 e^{-x} + C_2 e^{-2x}$

Ex. 3.3.13 *Solve* $\dfrac{d^4y}{dx^4} - 2\dfrac{d^2y}{dx^2} + y = 0.$

GTU : Winter-18, AEM, Marks 3

Sol. :

Here $y^{iv} - 2y'' + y = 0$

A.E. $D^4 - 2D^2 + 1 = 0$

$\therefore$ $(D^2 - 1)^2 = 0$

$\therefore$ $D^2 = 1$ two times

$\therefore$ $D = \pm 1$ two times

$\therefore$ $y = e^x (C_1 + C_2 x) + e^{-x}(C_3 + C_4 x)$

Ex. 3.3.14 : *Solve* $y'' + 6y' + 9y = 0,\ y(0) = 1,\ y'(0) = 2.$

GTU : Summer-23, Maths-2, Marks 4

Sol. : Here $y'' + 6y' + 9y = 0$

A.E. $D^2 + 6D + 9 = 0$

$$(D+3)^2 = 0$$

$$D = -3, -3$$

$\therefore$ $y = e^{-3x}(C_1 + C_2 x)$

Here, $y(0) = 1 \Rightarrow y = 1$ When $x = 0$

$$1 = C_1$$

$$y^1 = -3e^{-3x}(C_1 + C_2 x) + e^{-3x} C_2$$

Here, $y'(0) = 2 \Rightarrow y' = 2$ when $x = 0$

$$2 = -3C_1 + C_2 \Rightarrow 2 = -3 + C_1 \Rightarrow C_2 = 5$$

$\therefore$ $y = e^{-3x}(1 + 5x)$

Ex. 3.3.15 : *Solve $y''-5y'-14y=0$, $y(0)=3$, $y'(0)=1$.*

GTU : Summer-23, Maths-2, Marks 4

Sol. : Here $y''-5y'-14y=0$ is given

A.E. $$D^2-5D-14 = 0$$

$$D^2-7D+2D-14 = 0$$

$$D(D-7)+2(D-7) = 0$$

$$(D-7)(D+2) = 0$$

$\therefore$ $$D = 7, D = -2$$

$$y = C_1e^{7x}+C_2e^{-2x}$$

Here, $y(0) = 3 \Rightarrow y = 3$ when $x = 0$

$$3 = C_1+C_2 \quad ...(1)$$

Here, $y'(0) = 1 \Rightarrow y' = 1$ when $x = 0$

$$y' = 7C_1e^{7x}-2C_2e^{-2x}$$

$$1 = 7C_1-2C_2 \quad ...(2)$$

$\therefore$ $C_1+C_2=3$, $7C_1-2C_2=1$ | $2C_1+2C_2=6$, $7C_1-2C_2=1$

$$9C_1 = 7 \Rightarrow C_1 = 7/9$$

and $$7/9+C_2 = 3 \Rightarrow C_2 = 3-7/9 = \frac{27-7}{9} = \frac{20}{7}$$

$\therefore$ $$y = 7/9e^{7x}+20/7e^{-2x}$$

Ex. 3.3.16 : *Solve i) $(D^2+7D-18)y=0$*
ii) $y''+y'+2y=0$

GTU : Winter-23, Maths-2, Marks 4

Sol. : i) $(D^2+7D-18)y = 0$

A.E. $$D^2+7D-18 = 0$$

$$D^2+9D-2D-18 = 0$$

$$D(D+9)-2(D+9) = 0$$

$$(D-2)(D+9) = 0$$

$\therefore$ $$D = 2 \text{ and } D = -9$$

$\therefore$ $$y = C_1e^{2x}+C_2e^{-9x}$$

ii) $y''+y'+2y=0$

A.E. $$D^2+D+2 = 2$$

$a = 1, b = 1, c = ?$

$$\Delta = b^2-4ac = 1-8 = -7$$

$$D_1, D_2 = \frac{-b\pm\sqrt{\Delta}}{2a}$$

$$= \frac{-1\pm\sqrt{-7}}{2} = \frac{-1}{2}\pm\frac{\sqrt{7}}{2}i$$

$$y = e^{-1/2x}\left(C_1\cos\frac{\sqrt{7}}{2}x+C_2\sin\frac{\sqrt{7}}{2}x\right)$$

Ex. 3.3.17 : *Solve $2D^2y+Dy-6y=0$.*

GTU : Winter-22, Maths-2, Marks 3

Sol. : Here, $\left(2D^2+D-6\right)y = 0$

A.E. $$2D^2+D-6 = 0$$

$\therefore$ $$2D^2+4D-3D-6 = 0$$

$\therefore$ $$2D(D+2)-3(D+2) = 0$$

$$(2D-3)(D+2) = 0$$

$$2D = 3, D+2 = 0$$

$\Rightarrow$ $$D = 3/2, D = -2$$

$\therefore$ $$y = C_1e^{3/2}x+C_2e^{-2x}$$

3.4 Method of Undetermined Coefficients

GTU : Summer-18,19,20,21,22,23,24, Winter-18,19,21,20,21,22,23

Here we discuss technique for finding the particular integral of the linear differential equation with constant co-efficient.

$$\frac{dy^n}{dx^n}+C_1\frac{dy^{n-1}}{dx^{n-1}}+C_2\frac{dy^{n-2}}{dx^{n-2}}+\ldots+a_ny = r(x) \quad ... (3.4.1)$$

We know that the general solution of (3.4.1) can be written as $y(x) = y_C(x)+y_P(x)$.

where $y_c(x)$ is the complementary function of the homogenous part of equation (3.4.1).

To find the particular integral $y_P(x)$ of (3.4.1), we will use this method whenever the non-homogenous term is elementary function like, polynomial term (x^n), exponential term (e^{ax}) sine or cosine term (sin ax or cos ax), finite sum and/or product of these function.

Here, first we assume trail solution y_P depending on (R.H.S. term) or (r(x)) or non-homogenous terms as per the following Table. 3.4.1.

Sr. No.	R.H.S. at equations (3.4.1) Term or r(x)	Trail solution y_P
1.	$r(x) = e^{ax}$	$y_P = C \cdot e^{ax}$
2.	$r(x) = \sin ax$ or $\cos ax$ or $\sin ax + \cos ax$	$y_P = A \cos ax + B \sin ax$
3.	$r(x) = C_1 + C_2x + C_3x^2 + \ldots C_nx^n$ Example $r(x) = a + bx + cx^2 + dx^3$ $r(x) = a + bx + cx^2$ $r(x) = a + bx$ $r(x) = a$	$y_P = Ax^n + Bx^{n-1} + Cx^{n-2} + \ldots$ $y_P = Ax^3 + Bx^2 + Cx + D$ $y_P = Ax^2 + Bx + C$ $y_P = Ax + B$ $y_P = A$
4.	$r(x) = e^{ax} \sin bx$ or $e^{ax} \cos bx$	$y_P = e^{ax}(A \cos bx + B \sin bx)$
5.	$r(x) = xe^{ax}$ $r(x) = x^2 e^{ax}$	$y_P = e^{ax}(Ax + B)$ $y_P = e^{ax}(Ax^2 + Bx + C)$
6.	$r(x) = x \sin ax$ $r(x) = x^2 \sin ax$	$y_P = (Ax + B) \cos ax + (Cx + D) \sin ax$ $y_P = (Ax^2 + Bx + C) \cos ax + (Dx^2 + Ex + F) \sin ax$

Table 3.4.1

Additional rules for assuming trail solution y_P.

i) If no term of r(x) occurs in C.F. or y_C then suitable P.I. assumed from the above table.

ii) If any term of r(x) is also present in C.F. or y_C only once then multiple x corresponding to y_P, If any term of r(x) is also present in C.F. or y_C twice then multiply with x^2 corresponding to y_P term and so on.

iii) If r(x) = sin ax or cos ax or cos ax or addition of sin ax + cos ax then all term have same y_P.

Some example to assume trail solution y_P.

$$\text{i) If } r(x) = \left.\begin{matrix} e^{3x} \\ 2e^{3x} \\ 4e^{3x} \\ C \cdot e^{3x} \end{matrix}\right\} \rightarrow y_P = C \cdot e^{3x} \quad r(x) = \left.\begin{matrix} e^{bx} \\ 5e^{bx} \\ 6e^{bx} \\ C \cdot e^{bx} \end{matrix}\right\} \rightarrow y_P = C \cdot e^{bx}$$

$$\text{ii) } r(x) = \left.\begin{matrix} \sin 2x \\ \cos 2x \\ 2 \sin 2x \\ 3 \cos 2x \\ \sin 2x + \cos 2x \end{matrix}\right\} \rightarrow y_P = A \cos 2x + B \sin 2x$$

$$\text{iii) } r(x) = \left.\begin{matrix} ax^3 + bx^2 \\ 2x^3 + 3x^2 + x \\ x^3 + x + 1 \\ x^3 + x^2 + x + 1 \end{matrix}\right\} \rightarrow y_P = Ax^3 + Bx^2 + Cx + D$$

any 3^{rd} degree polynomial

Ex. 3.4.1 : *Solve the differential equation $(D^2+4)y = x^2 + \sin 2x$.* **GTU : Summer-18, Maths-3, Marks 3**

Sol. : Here, $(D^2+4)y = x^2+\sin 2x$ is given

$\therefore$ For C.F. $\quad D^2+4 = 0$

$\therefore \quad D^2 = -4 \Rightarrow D = \pm 2i$

Complex roots.

$\therefore \quad$ C.F. $= y_C = C_1 \cos 2x + C_2 \sin 2x$

Here, we use undeterminant co-efficient method to find P.I. (y_P)

$\therefore$ Let, the trail solution are,

$$y_P = Ax^2+Bx+C+D\cos 2x+E\sin 2x$$

Here sin 2x term appears once in y_C

$\therefore$ We multiply the term corresponding to sin 2x with x.

$$\therefore \quad y_P = Ax^2+Bx+C+Dx\cos 2x+Ex\sin 2x \quad \text{...(1)}$$

$$\therefore \quad y'_P = 2Ax+B+D\cos 2x-2Dx\sin 2x+E\sin 2x+2Ex\cos 2x$$

$$y''_P = 2A-2D\sin 2x-2D\sin 2x-4Dx\cos 2x+2E\cos 2x+2E\cos 2x-4Ex\sin 2x$$

$$y''_P = 2A-4D\sin 2x-4Dx\cos 2x-4E\cos 2x-4Ex\sin 2x$$

Substituting y'_P and y''_P in given equation.

$$y''_P+4y_P = x^2+\sin 2x$$

$$2A-4D\sin 2x-4Dx\cos 2x+4E\cos 2x-4Ex\sin 2x+4Ax^2+4Bx+4C+4Dx\cos 2x+4Ex\sin 2x = x^2+\sin 2x$$

$$\therefore \quad 4Ax^2+4Bx+(2A+4C)-4D\sin 2x+4E\cos 2x = x^2+\sin 2x$$

Comparing the corresponding co-efficient from both sides,

$$4A = 1 \Rightarrow A = \frac{1}{4}$$

$$4B = 0 \Rightarrow B = 0$$

$$2A+4C = 0 \Rightarrow 2\times\frac{1}{4}+4C = 0$$

$$\Rightarrow 4C = -\frac{1}{2}$$

$$\Rightarrow C = -\frac{1}{8}$$

$$-4D = 1 \Rightarrow D = -\frac{1}{4}$$

$$4E = 0 \Rightarrow E = 0$$

Hence, P.I. $\quad y_P = \frac{1}{4}x^2-\frac{1}{8}-\frac{1}{4}x\cdot\cos 2x$

$$\therefore \quad y = y_C+y_P$$

$$= C_2\cos 2x+C_2\sin 2x+\frac{1}{4}x^2-\frac{1}{8}-\frac{1}{4}x\cdot\cos 2x$$

Ex. 3.4.2 : *Using method of undeterminant co-efficient solve* $y'' - 2y' + y = e^x + x$.

GTU : Summer-18, Maths-3, Marks 3

Sol. : Here, $y'' - 2y' + y = e^x + x$ is given

$\therefore$ For C.F. we write.

$$D^2 - 2D + 1 = 0$$
$$(D-1)^2 = 0$$
$$D = 1, 1 \text{ (two times)}$$
$$\text{C.F.} = e^x(C_1 + C_2 x)$$

Now, for P.I. we use U.C. method

$\therefore$ Let us take, trail solution,

$$y_P = C_1 \cdot e^x + C_2 x + C_3$$

Here, e^x term appears two times in C.F. so, we multiply the term corresponding to e^x with x^2

$$\therefore\ y_P = C_1 x^2 e^x + C_2 x + C_3$$
$$\therefore\ y'_P = 2C_1 x e^x + C_1 x^2 e^x + C_2$$
$$y''_P = 2C_1 e^x + 2C_1 x e^x + 2C_1 x e^x + C_1 x^2 e^x$$

Now put y_P, y'_P and y''_P in given equation,

$$\therefore\ y''_P - 2y'_P + y_P = e^x + x$$
$$\therefore\ 2C_1 e^x + 4C_1 x e^x + C_1 x^2 e^x$$
$$-4C_1 x e^x - 2C_1 x^2 e^x - 2C_2 + C_1 e^x x^2 + C_2 x + C_3 = e^x + x$$
$$\therefore\ 2C_1 e^x + C_2 x + (C_3 - 2C_2) = e^x + x$$

Comparing the corresponding co-efficient from both sides.

$$\therefore\ 2C_1 = 1 \Rightarrow C_1 = 1/2$$
$$C_2 = 1$$
$$C_3 - 2C_2 = 0 \Rightarrow C_3 - 2 = 0 \Rightarrow C_3 = 2$$
$$\therefore\ y_P = \mathbf{\frac{1}{2} x^2 e^x + x + 2}$$

Ex. 3.4.3 : *Using method of undeterminant co-efficient solve* $y'' + 4y = 8x^2$.

GTU : Summer-23, AEM, Marks 4, Summer-19,21, Maths-3, Marks 3.5, Winter-21, AEM, Marks 4, Winter-23, AEM, Marks 3

Sol. : Here, $y'' + 4y = 8x^2$ is given

$$\therefore \text{A.E.}\quad D^2 + 4 = 0$$
$$D^2 = -4$$
$$\therefore\ D = \pm 2i$$
$$\text{C.F.} = C_1 \cos 2x + C_2 \sin 2x$$

Let the trail solution is,

$$y_P = Ax^2 + Bx + C$$
$$\therefore\ y'_P = 2Ax + B$$
$$y''_P = 2A$$

Put, y''_P and y_P in given equation,

$$\therefore\ y''_P + 4y_P = 8x^2$$
$$2A + 4Ax^2 + 4Bx + 4C = 8x^2$$
$$\therefore\ 4Ax^2 + 4Bx + (2A + 4C) = 8x^2$$

Comparing corresponding co-efficient on both sides, we get

$$4A = 8 \Rightarrow A = 2$$
$$4B = 0 \Rightarrow B = 0$$
$$2A + 4C = 0 \Rightarrow C = -1$$
$$\therefore\ y_P = 2x^2 - 1$$

Now, $y = y_C + y_P = \text{C.F.} + \text{P.I.}$

$$y = \mathbf{C_1 \cos 2x + C_2 \sin 2x + 2x^2 - 1}$$

Ex. 3.4.4 : *Using method of undetermined multipliers solve* $y'' - 3y' + 2y = e^x$.

GTU : Summer-19,21 Maths-3, Marks 3.5

Sol. : Here, $y'' - 3y' + 2y = e^x$ is given

$$\therefore \text{A.E.}\quad D^2 - 3D + 2 = 0$$
$$\therefore\ D^2 - 2D - D + 2 = 0$$
$$\therefore\ D(D-2) - 1(D-2) = 0$$
$$(D-1)(D-2) = 0$$
$$\therefore\ D = 1 \text{ and } D = 2$$
$$\therefore\ \text{C.F.} = y_C = \mathbf{C_1 e^x + C_2 e^{2x}}$$

Let the trail solution

$$y_P = C_1e^x$$

Here, e^x term appears in C.F. at once so, we multiply the term corresponding e^x with x.

$$\therefore \quad y_P = C_1xe^x$$

$$y'_P = C_1e^x + C_1xe^x$$

$$y''_P = C_1e^x + C_1e^x + C_1xe^x$$

$$\therefore \quad y''_P = 2C_1e^x + C_1xe^x$$

Now, put y_P, y'_P and y''_P in given equation

$$\therefore \quad y''_P - 3y'_P + 2y_P = e^x$$

$$\therefore 2C_1e^x + C_1xe^x - 3C_1e^x - 3C_1xe^x + 2C_1xe^x = e^x$$

$$-C_1e^x = e^x$$

$$\therefore \quad -C_1 = 1 \Rightarrow C_1 = -1$$

$$\therefore \quad y_P = -e^x \cdot x$$

Final solution :

$$y = y_C + y_P$$

$$= \mathbf{C_1e^x + C_2e^{2x} - e^x \cdot x}$$

Ex. 3.4.5 : *Solve* $y'' - 5y' + 6y = e^{4x}$.

GTU : Summer-20, Maths-3, Marks 3

Sol. : Here, $y'' - 5y' + 6y = e^{4x}$ is given

$$\therefore \text{ A.E.} \quad D^2 - 5D + 6 = 0$$

$$\therefore \quad D^2 - 3D - 2D + 6 = 0$$

$$D(D-3) - 2(D-3) = 0$$

$$\therefore \quad (D-2)(D-3) = 0$$

$$\therefore \quad D = 2 \text{ and } D = 3$$

$$\therefore \quad \text{C.F.} = y_C = C_1e^{2x} + C_2e^{3x}$$

Now, here we use U.C. method to find y_P(P.I.)

Let trail solution,

$$y_P = C_1e^{4x}$$

$$\therefore \quad y'_P = 4C_1e^{4x}$$

$$\therefore \quad y''_P = 16C_1e^{4x}$$

Put y_P, y'_P and y''_P in given equation.

$$\therefore \quad y''_P - 5y'_P + 6y_P = e^{4x}$$

$$16C_1e^{4x} - 20C_1e^{4x} + 6C_1e^{4x} = e^{4x}$$

$$2C_1e^{4x} = e^{4x}$$

$$\therefore \quad 2C_1 = 1 \Rightarrow C_1 = 1/2$$

$$\therefore \quad y_P = \text{P.I.} = \frac{1}{2}e^{4x}$$

Final solution : $y = y_C + y_P$

$$\therefore \quad y = \mathbf{C_1e^{2x} + C_2e^{3x} + \frac{1}{2}e^{4x}}$$

Ex. 3.4.6 : *Using the method of undetermined co-efficient, solve* $y''' + 3y'' + 2y' = x^2 + 4x + 8$.

GTU : Summer-20, Maths-3, Marks 7

Sol. : Here, $y''' + 3y'' + 2y' = x^2 + 4x + 8$ is given

$$\text{A.E.} \quad D^3 + 3D^2 + 2D = 0$$

$$\therefore \quad D(D^2 + 3D + 2) = 0$$

$$\therefore \quad D(D^2 + 2D + D + 2) = 0$$

$$\therefore \quad D(D + (D+2) + 1(D+2)) = 0$$

$$\therefore \quad D(D+2)(D+1) = 0$$

$$\therefore \quad D = 0, -1, -2$$

$$\therefore \quad \text{C.F.} = y_C = C_1 + C_2e^{-x} + C_3e^{-2x}$$

Now, we use U.C. method to find y_P(P.I.)

$\therefore$ Let us consider trail solution,

$$y_P = Ax^2 + Bx + C$$

Here, one of the solution basis 1 is appears in C.F. at once.

So, we multiple trail solution y_P with x.

$$\therefore \quad y_P = Ax^3 + Bx^2 + Cx$$

$$y'_P = 3Ax^2 + 2Bx + C$$

$$y''_P = 6Ax + 2B$$

$$y'''_P = 6A$$

Now, put y_P, y'_P, y''_P and y'''_P in given equation,

$$y'''_P + 3y''_P + 2y'_P = x^2 + 4x + 8$$

$$6A + 18Ax + 6B + 6Ax^2 + 4Bx + 2C = x^2 + 4x + 8$$

$$\therefore \quad 6Ax^2 + (18A + 4B)x + (6A + 6B + 2C) = x^2 + 4x + 8$$

Comparing both side of co-efficient,

$$\therefore \quad 6A = 1 \Rightarrow A = 1/6$$

$$18A + 4B = 4 \Rightarrow 18\left(\frac{1}{6}\right) + 4B = 4$$

$$\Rightarrow 3 + 4B = 4$$

$$\Rightarrow 4B = 1$$

$$\Rightarrow B = 1/4$$

$$6A + 6B + 2C = 8$$

$$6\left(\frac{1}{6}\right) + 6\left(\frac{1}{4}\right) + 2C = 8$$

$$1 + \frac{3}{2} + 2C = 8$$

$$\therefore \quad 2C = 7 - \frac{3}{2}$$

$$\therefore \quad 2C = \frac{11}{2} \Rightarrow C = \frac{11}{4}$$

$$\therefore \quad y_P = \frac{1}{6}x^3 + \frac{1}{4}x^2 + \frac{11}{4}x$$

$\therefore$ **Final solution :** $\quad y = y_C + y_P = \mathbf{C_1 + C_2 e^{-x} + C_3 e^{-2x} + \frac{1}{6}x^3 + \frac{1}{4}x^2 + \frac{11}{4}x}$

Ex. 3.4.7 : *Using the method of undetermined co-efficient solve $y'' - 2y' + 5y = 5x^3 - 6x^2 + 6x$.*

GTU : Summer-18, AEM, Marks 4, Summer-23, Maths-3, 7 Marks, Summer-24, Maths-2, Marks 7

Sol. : Here, $y'' - 2y' + 5y = 5x^3 - 6x^2 + 6x$

A.E. $\quad D^2 - 2D + 5 = 0$

Here, $\quad a = 1,\ b = -2$ and $c = 5$

$$\Delta = b^2 - 4ac = 4 - 4(1)(5) = 4 - 20 = -16$$

$$\therefore \quad D_1, D_2 = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{2 \pm \sqrt{-16}}{2} = \frac{2 \pm 4i}{2}$$

$$D_1, D_2 = 1 \pm 2i$$

$$\therefore \quad \text{C.F.} = y_C = e^x (C_1 \cos 2x + C_2 \sin 2x)$$

Let us consider trail solution,

$$y_P = Ax^3 + Bx^2 + Cx + D$$

$$\therefore \quad y'_P = 3Ax^2 + 2Bx + C$$

$$y''_P = 6Ax + 2B$$

Now, put y_P, y'_P and y''_P in given equation,

$$\therefore \quad y''_P - 2y'_P + 5y_P = 5x^3 - 6x^2 + 6x$$

$$6Ax + 2B - 6Ax^2 - 4Bx - 2C + 5Ax^3 + 5Bx^2 + 5Cx + 5D = 5x^3 - 6x^2 + 6x$$

$$\therefore 5Ax^3 + (5B - 6A)x^2 + (6A - 4B + 5C)x + (2B - 2C + 5D) = 5x^3 - 6x^2 + 6x$$

Comparing co-efficient both side,

$$5A = 5 \Rightarrow A = 1$$

$$5B - 6A = -6 \Rightarrow 5B - 6 = -6$$

$$\Rightarrow 5B = 0 \Rightarrow B = 0$$

$$6A - 4B + 5C = 6 \Rightarrow 6 - 4(0) + 5C = 6$$

$$\Rightarrow 6 + 5C = 6$$

$$\Rightarrow 5C = 0 \Rightarrow C = 0$$

$$2B - 2C + 5D = 0 \Rightarrow 2(0) - 2(0) + 5D = 0$$

$$\Rightarrow D = 0$$

$$\therefore \quad y_P = x^3 - 6x^2$$

$$\therefore \quad y = y_C + y_P$$

$$\mathbf{y = e^x (C_1 \cos 2x + C_2 \sin 2x) + x^3 - 6x^2}$$

Ex. 3.4.8 : *Solve $(D^2 - 4)y = e^x + \sin 2x$.* **GTU : Winter-18, Maths-3, Marks 3**

Sol. : Here, $(D^2 - 4)y = e^x + \sin 2x$ is given,

A.E. $$D^2 - 4 = 0 \Rightarrow D^2 = 4$$

$$\Rightarrow D = \pm 2$$

$$\therefore \quad y_C = C_1 e^{2x} + C_2 e^{-2x}$$

Here, we use U.C. method to find y_P

Let us consider trail solution,

$$y_P = C_1 e^x + C_2 \cos 2x + C_3 \sin 2x$$

$$y'_P = C_1 e^x - 2C_2 \sin 2x + 2C_3 \cos 2x$$

$$y''_P = C_1 e^x - 4C_2 \cos 2x - 4C_3 \sin 2x$$

Put y_P and y''_P in given equation.

$$\therefore \quad y''_P - 4y_P = e^x + \sin 2x$$

$$C_1 e^x - 4C_2 \cos 2x - 4C_3 \sin 2x - 4C_1 e^x - 4C_2 \cos 2x - 4C_3 \sin 2x = e^x + \sin 2x$$

$$\therefore -3C_1 e^x - 8C_2 \cos 2x - 8C_3 \sin 2x = e^x + \sin 2x$$

Comparing co-efficient both sides,

$$\therefore \quad -3C_1 = 1 \Rightarrow C_1 = -\frac{1}{3}$$

$$-8C_2 = 0 \Rightarrow C_2 = 0$$

$$-8C_3 = 1 \Rightarrow C_3 = -\frac{1}{8}$$

$$\therefore \quad y_P = -\frac{1}{3}e^x - \frac{1}{8}\sin 2x$$

$$\therefore \quad y = y_C + y_P$$

$$\mathbf{y = C_1 e^{2x} + C_2 e^{-2x} - \frac{1}{3}e^x - \frac{1}{8}\sin 2x}$$

Ex. 3.4.9 : *Using method of undetermined co-efficient method to solve $y'' + 2y' + 10y = 25x^2 + 3$.*

GTU : Winter-19, Maths-3, Marks 3

Sol. : Here, $y'' + 2y' + 10y = 25x^2 + 3$ is given,

A.E. $$D^2 + 2D + 10 = 0$$

Here, $$a = 1, b = 2, c = 10$$

$$\therefore \quad \Delta = b^2 - 4ac = 4 - 4(1)(10) = 4 - 40 = -36$$

$$\therefore \quad D_1, D_2 = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-2 \pm \sqrt{-36}}{2} = \frac{-2 \pm 6i}{2} = -1 \pm 3i$$

$$y_C = \text{C.F.} = e^{-x}(C_1 \cos 3x + C_2 \sin 3x)$$

Now, let us consider trail solution

$$y_P = Ax^2 + Bx + C$$

$$y'_P = 2Ax + B$$

$$\therefore \quad y''_P = 2A$$

Now, put y_P, y'_P and y''_P in given equation

$$y''_P + 2y'_P + 10y_P = 25x^2 + 3$$

$$2A + 4Ax + 2B + 10Ax^2 + 10Bx + 10C = 25x^2 + 3$$

Comparing co-efficient both side,

$$10Ax^2 + (4A + 10B)x + (2A + 2B + 10C) = 25x^2 + 3$$

$$\therefore \quad 10A = 25 \Rightarrow A = \frac{25}{10} = \frac{5}{2}$$

$$4A + 10B = 0 \Rightarrow 4\left(\frac{5}{2}\right) + 10B = 0$$

$$\Rightarrow 10 + 10B = 0$$

$$\Rightarrow 10\,B = -10 \Rightarrow \mathbf{B = -1}$$

$$2A + 2B + 10C = 3 \Rightarrow 2\left(\frac{5}{2}\right) + 2(-1) + 10\,C = 3$$

$$\Rightarrow 5 - 2 + 10C = 3$$

$$\Rightarrow 3 + 10C = 3$$

$$\Rightarrow 10C = 0 \Rightarrow \mathbf{C = 0}$$

$$\therefore \quad y_P = \frac{5}{2}x^2 - x$$

$$\therefore \quad y = y_C + y_P$$

$$\mathbf{y = e^{-x}(C_1 \cos\ 3x + C_2 \sin\ 3x) + \frac{5}{2}x^2 - x}$$

Ex. 3.4.10 : *Using method of undetermined co-efficient method to solve $y'' - 9y = x + e^{2x} - \sin 2x$.*

GTU : Winter-19, Maths-3, Marks 7

Sol. : Here , $y'' - 9y = x + e^{2x} - \sin\ 2x$ is given

$\therefore$ A.E. $\quad D^2 - 9 = 0 \Rightarrow D^2 = 9$

$$\Rightarrow D = \mp 3$$

$$\therefore \quad y_C = C_1 e^{3x} + C_2 e^{-3x}$$

Now, let us consider trail solution

$$y_P = Ax + B + C \cdot e^{2x} + D\cos 2x + E\sin 2x$$

$$y'_P = A + 2Ce^{2x} - 2D\sin 2x + 2E\cos 2x$$

$$\therefore \quad y''_P = 4Ce^{2x} - 4D\cos 2x - 4E\sin 2x$$

Now, put y_P and y''_P in given equation

$$\therefore \quad y''_P - 9y_p = x + e^{2x} - \sin 2x$$

$$\therefore \quad 4Ce^{2x} - 4D\cos\ 2x - 4E\sin\ 2x - 9Ax - 9B - 9Ce^{2x} - 9D\cos 2x - 9E\sin 2x = x + e^{2x} - \sin 2x$$

Comparing co-efficient both sides,

$$\therefore \quad -5C = 1 \Rightarrow C = -\frac{1}{5}$$

$$-9A = 1 \Rightarrow A = -\frac{1}{9}$$

$$-9B = 0 \Rightarrow B = 0$$

$$-13D = 0 \Rightarrow D = 0$$

$$-13E = -1 \Rightarrow E = \frac{1}{13}$$

$$\therefore \quad y_P = -1/9\,x - 1/5\,e^{2x} + 1/13 \sin 2x$$

$$\therefore \quad y = y_C + y_P$$

$$\therefore \quad y = C_1e^{3x} + C_2e^{-3x} - \frac{1}{9}x - \frac{1}{5}e^{2x} + \frac{1}{13}\sin 2x$$

Ex. 3.4.11 : *Solve the initial value problem $y''' + 3y'' + 3y' + y = 30\,e^{-x}$, $y(0) = 3$, $y'(0) = -3$. $y''(0) = 47$, by the method of undetermined co-efficients.*

GTU : Winter-21, Maths-3, Marks 7

Sol. :

Here , $y''' + 3y'' + 3y' + y = 30\,e^{-x}$ is given

A.E. $(D^3 + 3D^2 + 3D + 1) = 0$

$$\therefore \quad (D+1)^3 = 0$$

$$\therefore \quad D = -1, -1, -1$$

$$\therefore \quad y_C = e^{-x}(C_1 + C_2x + C_3x^2)$$

Now, let us consider trail solution

$$y_P = C \cdot e^{-x}$$

Here, e^{-x} appears three times in y_C So, we multiply x^3 with trail solution y_P

$$\therefore \quad y_P = Cx^3e^{-x}$$

$$y'_P = 3Cx^2e^{-x} - Cx^3e^{-x}$$

$$y''_P = 6\,Cxe^{-x} - 3Cx^2e^{-x} - 3Cx^2e^{-x} + Cx^3e^{-x}$$

$$y''_P = 6\,Cxe^{-x} - 6Cx^2e^{-x} + Cx^3e^{-x}$$

$$y'''_P = 6Ce^{-x} - 6Cxe^{-x} - 12Cxe^{-x} + 6Cx^2e^{-x} + 3Cx^2e^{-x} - Cx^3e^{-x}$$

$$\therefore \quad y'''_P = 6Ce^{-x} - 18Cxe^{-x} + 9Cx^2e^{-x} - Cx^3e^{-x}$$

Now, put y_P, y'_P, y''_P and y'''_P in given equation,

$$\therefore \quad y''' + 3y''_P + 3y'_P + y_P = 30e^{-x}$$

$\therefore\ 6Ce^{-x} - 18Cxe^{-x} + 9Cx^2e^{-x} - Cx^3e^{-x} + 18Cxe^{-x} - 18Cx^2e^{-x} + 3Cx^3e^{-x} + 9Cx^2e^{-x} - 3Cx^3e^{-x} + Cx^3e^{-x} = 30e^{-x}$

$$\therefore \quad 6Ce^{-x} = 30e^{-x}$$

$$\therefore \quad 6C = 30 \Rightarrow \mathbf{C = 5}$$

$$\therefore \quad y_P = 5x^3e^{-x}$$

$$\therefore \quad y = y_C + y_P = e^{-x}(C_1 + C_2x + C_3x^2) + 5x^3e^{-x}$$

$$\therefore \quad y(0) = 3 \Rightarrow y = 3 \text{ when } x = 0$$

$$\therefore \quad 3 = C_1$$

Now, $\quad y_1 = -e^{-x}(C_1+C_2x+C_3x^2)+e^{-x}(C_2+3C_3x)+15x^2e^{-x}-5x^3e^{-x}$

Here, $\quad y'(0) = -3 \Rightarrow y' = -3$ when $x = 0$

$$-3 = -(C_1)+(C_2)$$

$$\Rightarrow -3 = -3+C_2 \Rightarrow C_2 = 0$$

Now, $y'' =$

$e^{-x}(C_1+C_2x+C_3x^2)-e^{-x}(C_2+2C_3x)-e^{-x}(C_2+3C_3x)+e^{-x}(3C_3)+30xe^{-x}-15x^2e^{-x}-15x^2e^{-x}+5x^3e^{-x}$

Here, $\quad y''(0) = -47 \Rightarrow y'' = -47$ when $x = 0$

$$-47 = C_1-C_2-C_2+3C_3$$

$$-47 = 3-0-0+3C_3$$

$$-49 = 3C_3 \Rightarrow C_3 = -\frac{49}{3}$$

$$\therefore \quad \mathbf{y = e^{-x}(3-49/3\,x^2)+5x^3e^{-x}}$$

GTU : Winter-22, Maths-3, Marks 3

Ex. 3.4.12 : *Solve* $(D^2+5D+6)y = e^x$.

Sol. : Here, $(D^2+5D+6)y = e^x$ is given

A.E. $\quad D^2+5D+6 = 0$

$$D^2+3D+2D+6 = 0$$

$$D(D+3)+2(D+3) = 0$$

$$\therefore \quad (D+2)(D+3) = 0$$

$$\therefore \quad D = -2, D = -3$$

$$\therefore \quad y_C = C_1e^{-2x}+C_2e^{-3x}$$

Let us consider trail solution,

$$y_P = Ce^x$$

$$y'_P = C\cdot e^x$$

$$y''_P = C\cdot e^x$$

Put y_P, y'_P and y''_P in given equation,

$$y''_P+5y'_P+6y_P = e^x$$

$$\therefore \quad Ce^x+5Ce^x+6Ce^x = e^x$$

$$\therefore \quad 12Ce^x = e^x$$

$$\therefore \quad 12\,C = 1 \Rightarrow C = 1/12$$

$$\therefore \quad y_P = \frac{1}{12}e^x$$

$$\therefore \quad y = y_C+y_P = \mathbf{C_1e^{-2x}+C_2e^{-3x}+\frac{1}{12}e^x}$$

Ex. 3.4.13 : *Using the method of undetermined co-efficient, find the general solution of the differential equation* $y'' + 9y = 2x^2$

GTU : Winter-22, Maths-3, Winter-23, Maths-2, Marks 7

Sol. : Here, $y'' + 9y = 2x^2$

$\therefore$ A.E. $D^2 + 9 = 0 \Rightarrow D = \pm 3i$

$$\therefore \quad y_C = C_1 \cos 3x + C_2 \sin 3x$$

Now, let us consider trail solution y_P

$$\therefore \quad y_P = Ax^2 + Bx + C$$

$$y'_P = 2A\,x + B$$

$$y''_P = 2A$$

Put y_P, y'_P and y''_P in given equation,

$$\therefore \quad y''_P + 9y_P = 2x^2$$

$$2A + 9Ax^2 + 9Bx + 9C = 2x^2$$

$$\therefore \quad 9Ax^2 + 9Bx + (2A + 9C) = 2x^2$$

Comparing co-efficient both sides,

$$\therefore \quad 9A = 2 \Rightarrow A = \frac{2}{9}$$

$$9B = 0 \Rightarrow B = 0$$

$$2A + 9C = 0 \Rightarrow 2\left(\frac{2}{9}\right) + 9C = 0$$

$$\Rightarrow \frac{4}{9} + 9C = 0 \Rightarrow 9C = -\frac{4}{9}$$

$$\Rightarrow C = -\frac{4}{81}$$

$$\therefore \quad y_P = \frac{2}{9}x^2 - \frac{4}{81}$$

$$\therefore \quad y = y_C + y_P$$

$$\mathbf{y = C_1 \cos 3x + C_2 \sin 3x + \frac{2}{9}x^2 - \frac{4}{81}}$$

Ex. 3.4.14 : *Solve* $y'' - 3y' + 2y = e^{3x}$

GTU : Summer-18, AEM, Marks 3

Sol. : Here, $y'' - 3y' + 2y = e^{3x}$ is given

$\therefore$ A.E. $D^2 - 3D + 2 = 0$

$$(D-2)(D-1) = 0$$

$$\therefore \quad D = 2 \text{ and } D = 1$$

$$y_C = C_1 e^{2x} + C_2 e^{x}$$

Now, let us consider trail solution,

$$y_P = C e^{3x}$$

$$y'_P = 3C e^{3x},$$

$$y''_P = 9C e^{3x}$$

Put y_P, y'_P and y''_P in given equation.

$$y''_P - 3y'_P + 2y_P = e^{3x}$$

$$9Ce^{3x} - 9Ce^{3x} + 2Ce^{3x} = e^{3x}$$

$$\therefore \quad 2Ce^{3x} = e^{3x} \Rightarrow 2C = 1 \Rightarrow C = 1/2$$

$$\therefore \quad y_P = \frac{1}{2}e^{3x}$$

$$\therefore \quad y = y_C + y_P$$

$$\mathbf{y = C_1 e^{2x} + C_2 e^{x} + \frac{1}{2}e^{3x}}$$

Ex. 3.4.15 : *Solve* $(D^2 - D - 2)y = \sin 2x$.

GTU : Summer-20, AEM, Marks 4

Sol. : Here, $(D^2 - D - 2)\,y = \sin 2x$ is given

$\therefore$ A.E. $D^2 - D - 2 = 0$

$$D^2 - 2D + D - 2 = 0$$

$$(D-2)(D+1) = 0$$

$$\therefore \quad D = 2, -1$$

$$\therefore \quad y_C = C_1 e^{2x} + C_2 e^{-x}$$

Let us consider trail solution y_P

$$\therefore \quad y_P = A\cos 2x + B\sin 2x$$

$$y'_P = -2A\sin 2x + 2B\cos 2x$$

$$y''_P = -4A\cos 2x - 4B\sin 2x$$

Put, y_P, y'_P and y''_P in given equation

$$\therefore \quad y''_P - y'_P - 2y_P = \sin 2x$$

$$\therefore \quad -4A\cos 2x - 4B\sin 2x + 2A\sin 2x$$

$$-2B\cos 2x - 2A\cos 2x - 2B\sin 2x = \sin 2x$$

$$\therefore \quad (-6A - 2B)\cos 2x + (2A - 6B)\sin 2x = \sin 2x$$

Comparing co-efficient both sides.

$$\therefore \quad -6A-2B = 0 \qquad \therefore \quad -6A-2B = 0$$

$$2A-6B = 1 \qquad \underline{6A-18B = 1}$$

$$-20B = 3$$

$$\Rightarrow \quad B = -3/20$$

$$\therefore \quad -6A+\frac{6}{20} = 0$$

$$-6A = -\frac{6}{20}$$

$$\Rightarrow \quad \mathbf{A = \frac{1}{20}}$$

$$\therefore \quad y_P = \frac{1}{20}\cos 2x-\frac{3}{20}\sin 2x$$

$$\therefore \quad y = y_C+y_P = \mathbf{C_1e^{2x}+C_2e^{-x}+\frac{1}{20}\cos 2x-\frac{3}{20}\sin 2x}$$

Ex. 3.4.16 : *Using method of undetermined co-efficient, solve $y''-2y' = e^x \sin x$.* **GTU : Summer-22,24, Marks 7**

Sol. : Here, $y''-2y' = e^x \sin x$ is given

$$\therefore \text{ A.E.} \quad D^2-2D = 0$$

$$D(D-2) = 0$$

$\therefore$ D = 0 and D = 2

$$\therefore \quad y_C = \text{C.F.} = C_1e^{0x}+C_2e^{2x}$$

$$\text{C.F.} = C_1+C_2e^{2x}$$

Let trail solution of y'_P

$$y_P = e^x(A\cos x+B\sin x)$$

$$y'_P = e^x(-A\sin x+B\cos x)+e^x(A\cos x+B\sin x)$$

$$y''_P = e^x(-A\sin x+B\cos x)+e^x(-A\cos x-B\sin x)+e^x(A\cos x+B\sin x)+e^x(-A\sin x+B\cos x)$$

$$\therefore \quad y''_P = 2e^x(-A\sin x+B\cos x)$$

$$y''_P = -2Ae^x\sin x+2Be^x\cdot\cos x$$

Now, put y'_P and y''_P in given equation.

$$y''_P-2y'_P = e^x\cdot\sin x$$

$$-2Ae^x\sin x+2Be^x\cdot\cos x+2Ae^x\sin x-2Be^x\cdot\cos x-2Ae^x\cdot\cos x-2Be^x\sin x = e^x\cdot\sin x$$

$$-2Ae^x\cdot\cos x-2Be^x\cdot\sin = e^x\cdot\sin x$$

$$-2A = 0, \quad -2B = 1$$

$$A = 0 \text{ and } B = -\frac{1}{2}$$

$$\therefore \quad y_P = e^x\left(0\cdot\cos x-\frac{1}{2}\sin x\right) = -\frac{1}{2}e^x\sin x$$

$$\therefore \quad y_P = y_C+y_P = \mathbf{C_1+C_2e^{2x}-\frac{1}{2}e^x\cdot\sin x}$$

Ex. 3.4.17 : *Solve $y''+y' = 5e^x - \sin 2x$.* **GTU : Summer-19, AEM, Marks 4**

Sol. : Here, $y'' + y' = 5e^x - \sin 2x$ is given,

A.E. $$D^2+D = 0$$

$$\therefore \quad D(D+1) = 0$$

$$\therefore \quad D = 0 \text{ and } D = -1$$

$$\therefore \quad y_C = C_1e^{0x}+C_2e^{-x}$$

$$\therefore \quad y_C = C_1+C_2e^{-x}$$

Let us consider trail solution y_P

$$\therefore \quad y_P = C\cdot e^x + A\cos 2x + B\sin 2x$$

$$y_P' = C\cdot e^x - 2A\sin 2x + 2B\cos 2x$$

$$y_P'' = C\cdot e^x - 4A\cos 2x + 4B\sin 2x$$

Put, y_P, y_P' and y_P'' in given equation.

$$y_P''+y_P' = 5e^x - \sin 2x$$

$$C\cdot e^x - 4A\cos 2x - 4B\sin 2x + C\cdot e^x - 2A\sin 2x + 2B\cos 2x = 5e^x - \sin 2x$$

$$\therefore \quad 2C\cdot e^x + (2B-4A)\cos 2x + (-2A-4B)\sin 2x = 5e^x - \sin 2x$$

Comparing co-efficient both sides,

$$2C = 5 \Rightarrow C = 5/2$$

$$2B - 4A = 0 \qquad \therefore \quad -4A + 2B = 0$$

$$-2A - 4B = -1 \qquad\qquad -4A - 8B = -2$$

$$\underline{\quad + \quad + \quad + \quad}$$

$$10B = 2$$

$$\therefore \quad 2\left(\frac{1}{5}\right)-4A = 0 \qquad \therefore \quad B = \frac{1}{5}$$

$$\therefore \quad 4A = \frac{2}{5} \quad \Rightarrow \quad \mathbf{A = \frac{1}{10}}$$

$$\therefore \quad y_P = \frac{5}{2}e^x + \frac{1}{10}\cos 2x + \frac{1}{5}\sin 2x$$

$$\therefore \quad y = y_C+y_P$$

Ex. 3.4.18 : *Solve* $y'' - 2y' + y = 10 \cdot e^x$ **GTU : Summer-22, AEM, Marks 4, Winter-22, Maths-2, Marks 4**

Sol. : Here, $y'' - 2y' + y = 10.e^x$ is given

A.E. $D^2 - 2D + 1 = 0$

$\therefore \quad (D-1)^2 = 0$

$\therefore \quad D = 1, 1$

$\therefore \quad y_C = e^x(C_1 + C_2 x)$

Now, consider trail solution of y_P

$$y_P = C \cdot e^x$$

Here, e^x term appears twice in y_C

$\therefore$ We multiply x^2 in corresponding terms,

$\therefore \quad y_P = C \cdot x^2 e^x$

$y'_P = 2Ce^x x + Cx^2 e^x$

$y''_P = 2C e^x + 2C x e^x + 2C x e^x + Cx^2 e^x$

$\therefore \quad y''_P = 2Ce^x + 4Cxe^x + Cx^2 e^x$

Now, put y_P, y'_P and y''_P in given equation,

$y''_P - 2y'_P + y_P = 10e^x$

$\therefore \quad 2Ce^x + 4Cxe^x + Cx^2e^x - 4Cxe^x - 2Cx^2e^x + Cx^2e^x = 10e^x$

$\therefore \quad 2Ce^x = 10e^x \quad \therefore \quad 2C = 10 \Rightarrow C = 5$

$\therefore \quad y_P = 5x^2 e^x$

$\therefore \quad y = y_C + y_P = \mathbf{e^x(C_1 + C_2 x) + 5x^2 e^x}$

Ex. 3.4.19 : *Find the solution of* $y'' + 4y = 2 \sin 3x$ *by the method of undetermined co-efficients.* **GTU : Summer-23, AEM, Marks 7, Winter-22, Maths-2, Marks 7**

Sol. : Here, $y'' + 4y = 2 \sin 3x$ is given,

A.E. $D^2 + 4 = 0$

$\therefore \quad D^2 = -4$

$\therefore \quad D = \pm 2i$

$\therefore \quad y_C = C_1 \cos 2x + C_2 \sin 2x$

Let us consider trail solution y_P

$\therefore \quad y_P = A \cos 3x + B \sin 3x$

$y'_P = -3A \sin 3x + 3B \cos 3x$

$y''_P = -9A \cos 3x - 9B \sin 3x$

Put, y_P and y''_P in given equation,

$y''_P + 4y_P = 2 \sin 3x$

$\therefore \quad -9A \cos 3x - 9B \sin 3x + 4A \cos 3x + 4B \sin 3x = 2 \sin 3x$

$\therefore \quad -5A \cos 3x - 5B \sin 3x = 2 \sin 3x$

Comparing co-efficient both sides,

$\therefore \quad -5A = 0 \Rightarrow A = 0$

$-5B = 2 \Rightarrow B = -\frac{2}{5}$

$\therefore \quad y_P = -\frac{2}{5} \sin 2x$

$\therefore \quad y = y_C + y_P$

$= C_1 \cos 2x + C_2 \sin 2x - \frac{2}{5} \sin 2x$

Ex. 3.4.20 : *Using the method of undetermined co-efficient find the general solution of* $y'' - 4y' - 12y = 8x^2$. **GTU : Summer-24, AEM, Marks 7**

Sol. : Here, $y'' - 4y' - 12y = 8x^2$ is given.

$\therefore$ A.E. $\quad D^2 - 4D - 12 = 0$

$D^2 - 6D + 2D - 12 = 0$

$\therefore \quad D(D-6) + 2(D-6) = 0$

$\therefore \quad (D+2)(D-6) = 0$

$\therefore \quad D = -2, D = 6$

$\therefore \quad y_C = C_1 e^{-2x} + C_2 e^{6x}$

Now, let us consider trail solution y_P

$\therefore \quad y_P = Ax^2 + Bx + C$

$y'_P = 2Ax + B$

$y''_P = 2A$

Put, y_P, y'_P and y''_P in given equation.

$$\therefore \quad y_P'' - 4y_P' - 12y_P = 8x^2$$

$$\therefore 2A - 8Ax - 4B - 12Ax^2 - 12Bx - 12C = 8x^2$$

$$\therefore -12Ax^2 + (-8A - 12B)x + (2A - 4B - 12C) = 8x^2$$

Comparing co-efficient both sides.

$$\therefore \quad -12A = 8 \Rightarrow A = -\frac{8}{12} \Rightarrow A = -\frac{2}{3}$$

$$-8A - 12B = 0 \Rightarrow (-8)\left(-\frac{2}{3}\right) - 12B = 0$$

$$\Rightarrow 12B = \frac{16}{3} \Rightarrow B = \frac{16}{3\times 12} = \frac{4}{9}$$

$$\therefore \quad 2A - 4B - 12C = 0$$

$$\therefore \quad 2\left(-\frac{2}{3}\right) - 4\left(\frac{4}{9}\right) - 12C = 0$$

$$-\frac{4}{3} - \frac{16}{9} - 12C = 0$$

$$\therefore \quad -\frac{84}{27} = 12C$$

$$\therefore \quad C = -\frac{7}{27}$$

$$\therefore \quad y_P = -\frac{2}{3}x^2 + \frac{4}{9}x - \frac{7}{27}$$

$$\therefore \quad y = y_C + y_P$$

$$\therefore \quad \mathbf{y = C_1e^{-2x} + C_2e^{6x} - \frac{2}{3}x^2 + \frac{4}{9}x - \frac{7}{27}}$$

Ex. 3.4.21 : *Solve* $(D^2 - 4)y = 1 + e^x$.

GTU : Winter-18, AEM, Marks 4

Sol. : $(D^2 - 4)y = 1 + e^x$ is given

$$\therefore \quad \text{A.E. } D^2 - 4 = 0$$

$$D^2 = 4$$

$$\therefore \quad D = \pm 2$$

$$\therefore \quad y_C = C_1e^{2x} + C_2e^{-2x}$$

For y_P, we use U.C. method

Let us take trail solution y_P

$$\therefore \quad y_P = C + C_1e^x$$

$$y_P' = C_1e^x \text{ and } y_P'' = C_1e^x$$

Put y_P, y_P' and y_P'' in given equation,

$$y_P'' - 4y_P = 1 + e^x$$

$$C_1e^x - 4C - 4C_1e^x = 1 + e^x$$

$$-4C - 3C_1e^x = 1 + e^x$$

Comparing co-efficient both side,

$$-4C = 1 \Rightarrow C = -\frac{1}{4}$$

$$-3C_1 = 1 \Rightarrow C_1 = -\frac{1}{3}$$

$$\therefore \quad y_P = -\frac{1}{4} - \frac{1}{3}e^x$$

$$y = y_C + y_P$$

$$\therefore \quad \mathbf{y = C_1e^{2x} + C_2e^{-2x} - \frac{1}{4} - \frac{1}{3}e^x}$$

Ex. 3.4.22 : *Solve by the method of undetermined co-efficient* $y'' + 10y' + 25y = e^{-5x}$.

GTU : Winter-18, AEM, Marks 4

Sol. : Here, $y'' + 10y' + 25y = e^{-5x}$ is given,

$$\therefore \text{ A.E. } \quad D^2 + 10D + 25 = 0$$

$$\therefore \quad (D+5)^2 = 0$$

$$\therefore \quad D = -5, -5$$

$$\text{C.F.} = y_C$$

$$= e^{-5x}(C_1 + C_2x)$$

Let us consider trail solution y_P

$$\therefore \quad y_P = C\cdot e^{-5x}$$

Here, e^{-5x} appears twice in y_C

$\therefore$ We multiply x^2 in corresponding term y_P.

$$\therefore \quad y_P = Cx^2e^{-5x}$$

$$\therefore \quad y_P' = 2Cxe^{-5x} - 5Cx^2e^{-5x}$$

$$y_P'' = 2Ce^{-5x} - 10Cxe^{-5x} - 10Cxe^{-5x} + 25Cx^2e^{-5x}$$

$$\therefore \quad y_P'' = 2Ce^{-5x} - 20Cxe^{-5x} + 25Cx^2e^{-5x}$$

Now, put y_P, y_P' and y_P'' in given equation.

$$y'' + 10y_P' + 25y_P = e^{-5x}$$

$$2Ce^{-5x} - 20Cxe^{-5x} + 25Cx^2e^{-5x} + 20Cxe^{-5x} - 50Cx^2e^{-5x} + 25Cx^2e^{-5x} = e^{-5x}$$

$$\therefore \quad 2Ce^{-5x} = e^{-5x}$$

$$\therefore \quad 2C = 1 \Rightarrow C = 1/2$$

$$\therefore \quad y_P = \frac{1}{2}x^2e^{-5x}$$

$$\therefore \quad y = y_C + y_P = e^{-5x}(C_1 + C_2x) + \frac{1}{2}x^2e^{-5x}$$

Ex. 3.4.23 : *Using undetermined co-efficient method solve the differential equation* $y'' + y' - 6y = 6x + 3x^2 - 6x^3$.

GTU : Winter-19, AEM, Marks 7

Sol. : Here, $y'' + y' - 6y = 6x + 3x^2 - 6x^3$ is given

$$\therefore \text{ A.E.} \quad D^2 + D - 6 = 0$$

$$\therefore \quad D^2 + 3D - 2D - 6 = 0$$

$$D(D+3) - 2(D+3) = 0$$

$$(D-2)(D+3) = 0$$

$$D = 2 \text{ and } D = -3$$

$$\therefore \quad \text{C.F.} = y_C = C_1e^{2x} + C_2e^{-3x}$$

Let us consider trail solution y_P

$$y_P = Ax^3 + Bx^2 + Cx + D$$

$$y'_P = 3Ax^2 + 2Bx + C$$

$$y''_P = 6Ax + 2B$$

Put y_P, y'_P and y''_P in given equation,

$$\therefore \quad y''_P + y'_P - 6y_P = 6x + 3x^2 - 6x^3$$

$$(6Ax + 2B) + (3Ax^2 + 2Bx + C) - 6(Ax^3 + Bx^2 + Cx + D) = 6x + 3x^2 - 6x^3$$

$$\therefore \quad -6Ax^3 + (3A - 6B)x^2 + (6A + 2B - 6C)x + (2B + C - 6D) = 6x + 3x^2 - 6x^3$$

Comparing co-efficient both sides,

$$\therefore \quad -6A = -6 \Rightarrow A = 1$$

$$3A - 6B = 3 \Rightarrow 4 - 6B = 3 \Rightarrow B = 0$$

$$6A + 2B - 6C = 6$$

$$\Rightarrow \quad 6(1) + 2(0) - 6C = 6$$

$$\Rightarrow \quad C = 0$$

$$2B + C - 6D = 0$$

$$2(0) + (0) - 6D = 0$$

$$D = 0$$

$\Rightarrow$

$$y_P = x^3$$

$\therefore$

$$y = y_C + y_P = \mathbf{C_1e^{2x} + C_2\,e^{-3x} + x^3}$$

Ex. 3.4.24 : *Solve $y'' + 5y' + 6y = 12e^x$.*

GTU : Winter-21, AEM, Marks 4

Sol. : Here, $y'' + 5y' + 6y = 12\,e^x$ is given

$\therefore$ A.E. $\quad D^2 + 5D + 6 = 0$

$$(D+3)(D+2) = 0$$

$$D = -3, D = -2$$

$$\text{C.F.} = y_C = C_1e^{-3x} + C_2e^{-2x}$$

Now, for y_P we use U.C. method

Let us consider trail solution y_P

$$y_P = Ce^x$$

$$y'_P = C\cdot e^x$$

$$y''_P = C\cdot e^x$$

Put y_P, y'_P and y''_P in given equation,

$\therefore \quad y''_P + 5y'_P + 6y_P = 12\,e^x$

$$Ce^x + 5Ce^x + 6C\,e^x = 12\,e^x$$

$$12C\,e^x = 12\,e^x$$

$\therefore \quad 12\,C = 12 \Rightarrow \mathbf{C = 1}$

$\therefore \quad y_P = e^x$

$$y = y_C + y_P = \mathbf{C_1e^{-3x} + C_2e^{-2x} + e^x}$$

Ex. 3.4.25 : *Solve $y'' + 6y' + 9y = e^{3x}$.*

GTU : Winter-21, AEM, Marks 3

Sol. : Here, $y'' + 6y' + 9y = e^{3x}$ is given,

$\therefore$ A.E. $\quad D^2 + 6D + 9 = 0$

$$(D+3)^2 = 0$$

$\therefore \quad D = -3, -3$

$$\text{C.F.} = y_C = e^{-3x}(C_1 + C_2x)$$

Let us consider trail solution y_P

$\therefore \quad y_P = C\cdot e^{3x}$

$$y'_P = 3Ce^{3x}$$

$$y''_P = 9Ce^{3x}$$

Put y_P, y'_P and y''_P in given equation.

$\therefore \quad y''_P + 6y'_P + 9y_P = e^{3x}$

$$9Ce^{3x} + 18Ce^{3x} + 9C\,e^{3x} = e^{3x}$$

$$36Ce^{3x} = e^{3x}$$

$\therefore \quad 36\,C = 1 \Rightarrow \mathbf{C = \frac{1}{36}}$

$\therefore \quad y_P = \frac{1}{36}e^{3x}$

$\therefore \quad y = y_C + y_P = \mathbf{e^{-3x}(C_1 + C_2x) + \frac{1}{36}e^{3x}}$

Ex. 3.4.26 : *Solve $(D^3 - D^2 - 6D)y = x^2 + 1$.*

GTU : Winter-22, AEM, Marks 4

Sol. : Here, $(D^3 - D^2 - 6D)y = x^2 + 1$ is given

$\therefore$ A.E. $\quad (D^3 + D^2 + 6D)y = 0$

$$D(D^2 - D - 6) = 0$$

$\therefore \quad D(D^2 - 3D + 2D - 6) = 0$

$$D(D-3)(D+2) = 0$$

$$D = 0, 3, -2$$

$\therefore \quad \text{C.F.} = y_C = \mathbf{C_1 + C_2e^{3x} + C_3e^{-2x}}$

Here, we use U.C. Method to find y_P

$\therefore$ Ler us consider trail solution y_P

$$y_P = Ax^2 + Bx + C$$

Here term constant appears once in y_C

$\therefore$ Multiply x with y_P

$\therefore \quad y_P = Ax^3 + Bx^2 + Cx$

$$y'_P = 3Ax^2 + 2Bx + C$$

$$y''_P = 6Ax + 2B$$

$$y'''_P = 6A$$

put y'_P, y''_P and y'''_P in given equation.

$\therefore \; y'''_P - y''_P - 6y'_P = x^2 + 1$

$$6A - 6Ax - 2B - 18Ax^2 - 12Bx - 6C = x^2 + 1$$

$$-18Ax^2 + (-6A - 12B)x + (6A - 2B - 6C) = x^2 + 1$$

Comparing co-efficient both sides.

$$-18A = 1 \Rightarrow A = -\frac{1}{18}$$

$$-6A - 12B = 0 \Rightarrow B = 0$$

$$6A - 2B - 6C = 1$$

$$6\left(-\frac{1}{18}\right) - 2(0) - 6C = 1$$

$$-\frac{1}{3} - 6C = 1$$

$$\Rightarrow \quad 6C = -\frac{1}{3} - 1 = -\frac{4}{3}$$

$$\Rightarrow \quad C = -\frac{4}{18}$$

$$\therefore \quad y_P = -\frac{1}{18}x^3 - \frac{4}{18}$$

$$\therefore \quad y = y_C + y_P$$

$$\mathbf{y = C_1 + C_2e^{3x} + C_3e^{-2x} - \frac{1}{18}x^3 - \frac{4}{18}}$$

Ex. 3.4.27 : *Use the method of undetermined co-efficient to solve differential equation $y'' - 2y' + y = x^2e^x$.*

GTU : Summer-19, 20, Maths-2, Marks 7, Summer-23, Maths-2, Marks 3.5

Sol. : Here, $y'' - 2y' + y = x^2e^x$ is given

A.E. $\quad D^2 - 2D + 1 = 0$

$$(D-1)^2 = 0$$

$$D = 1, 1$$

$\therefore$ C.F. $\quad y_C = (C_1 + C_2x)e^x$

Let us consider trail solution y_P

$$\therefore \quad y_P = e^x(Ax^2 + Bx + C)$$

Now, e^x term appears twice in y_C multiply x^2 in y_P

$$\therefore \quad y_P = e^x(Ax^4 + Bx^3 + Cx^2)$$

$$\therefore \quad y'_P = e^x(Ax^4 + Bx^3 + Cx^2) + e^x(4Ax^3 + 3Bx^2 + 2Cx)$$

$$y''_P = e^x(Ax^4 + Bx^3 + Cx^2) + e^x(4Ax^3 + 3Bx^2 + 2Cx) + e^x(4Ax^3 + 3Bx^2 + 2Cx) + e^x(12Ax^2 + 6Bx + 2C)$$

$$\therefore \quad y''_P = e^x(Ax^4 + Bx^3 + Cx^2) + 2e^x(4Ax^3 + 3Bx^2 + 2Cx) + e^x(12Ax^2 + 6Bx + 2C)$$

Put y_P, y'_P and y''_P in given equation,

$$y''_P - 2y'_P + y_P = x^2e^x$$

$$e^x(Ax^4 + Bx^3 + Cx^2) + 2e^x(4Ax^3 + 3Bx^2 + 2Cx) + e^x(12Ax^2 + 6Bx + 2C)$$

$$-2e^x(Ax^4 + Bx^3 + Cx^2) - 2e^x(4Ax^3 + 3Bx^2 + 2Cx) + e^x(Ax^4 + Bx^3 + Cx^2) = x^2e^x$$

$$e^x(12Ax^2 + 6Bx + 2C) = x^2e^x$$

Now compare co-efficient both side,

$12A = 1,\ 6B = 0,\ 2C = 0$

$A = \frac{1}{12},\ B = 0,\ C = 0$

$$\therefore \quad y_P = \frac{1}{12} e^x x^4$$

$$\therefore \quad y = y_C + y_P$$

$$y = (C_1 + C_2 x)\, e^x + \frac{1}{12} e^x x^4$$

Ex. 3.4.28 : *Solve $y'' - 2y' + y = \cos 3x$.* **GTU : Summer-22, Maths-2, Marks 3**

Sol. : Here, $y'' - 2y' + y = \cos 3x$ is given

A.E.
$$D^2 - 2D + 1 = 0$$
$$(D-1)^2 = 0$$
$$D = 1, 1$$
$$\text{C.F.} = y_C\ e^x (C_1 + C_2 x)$$

Now, let trail solution y_P

$$y_P = A \cos 3x + B \sin 3x$$
$$y_P' = -3A \sin 3x + 3B \cos 3x$$
$$y_P'' = -9A \cos 3x - 9B \sin 3x$$

Put, y_P, y_P' and y_P'' in given equation

$$y_P'' - 2y_P' + y = \cos 3x$$

$-9A \cos 3x - 9B \sin 3x + 6A \sin 3x - 6B \cos 3x + A \cos 3x + B \sin 3x = \cos 3x$

$-8A \cos 3x - 8B \sin 3x + 6A \sin 3x - 6B \cos 2x = \cos 3x$

$\therefore (-8A - 6B) \cos 3x + (6A - 8B) \sin 3x = \cos 3x$

$\therefore (-8A - 6B) \cos 3x + (6A - 8B) \sin 3x = \cos 3x$

Comparing co-efficient both sides,

$$-8A - 6B = 1$$
$$6A - 8B = 0$$
$$\therefore \quad -48A - 36B = 6$$
$$48A - 64B = 0$$
$$-100\,B = 6 \Rightarrow B - \frac{6}{100}$$

$6A - 8\left(-\frac{3}{50}\right) = 0 \quad \Rightarrow \quad B = -\frac{3}{50}$

$$6A + \frac{24}{50} = 0$$

$$\therefore \quad 6A = -\frac{24}{50} \Rightarrow A = -\frac{24}{6\times 50}$$

$$\Rightarrow \quad A = -\frac{4}{50}$$

$$\therefore \quad y_P = -\frac{4}{50}\cos\ 3x - \frac{3}{50}\sin\ 3x$$

$$\therefore \quad y_P = y_C + y_P$$

$$y = e^x(C_1 + C_2 x) - \frac{4}{50}\cos 3x - \frac{3}{50}\sin 3x$$

Ex. 3.4.29 : *Using method of undetermined co-efficient obtain the solution of $y'' + 4y = \sin x$.*

GTU : Summer-23, Maths-2, Marks 4

Sol. : Here, $y'' + 4y = \sin x$

A.E. $\quad D^2 + 4 = 0 \Rightarrow D^2 = -4 \Rightarrow D = \pm 2i$

$$y_C = C_1 \cos\ 2x + C_2 \sin\ 2x$$

Now, let us consider trail solution y_P

$$y_P = A\cos x + B\sin x$$

$$y'_P = -A\sin x + B\cos x$$

$$y''_P = -A\cos x - B\sin x$$

Put y_P, y'_P and y''_P in given equation

$$y''_P + 4y_P = \sin x$$

$$-A\cos x - B\sin x + 4A\cos x + 4B\sin x = \sin x$$

$$3A\cos x + 3B\sin x = \sin x$$

$$\therefore \quad 3A = 0 \Rightarrow A = 0$$

$$3B = 1 \Rightarrow B = y_3$$

$$\therefore \quad y_P = \frac{1}{3}\sin x,$$

$$y = y_C + y_P = C_1\cos 2x + C_2\ \sin 2x + \frac{1}{3}\sin x$$

Ex. 3.4.30 : *Solve $(D^2 + 9)y = 2\sin 3x + \cos 3x$.*

GTU : Summer-24, Maths 2, Marks 4

Sol. : Here, $(D^2 + 9)y = 2\sin 3x + \cos 3x$

A.E. $\quad D^2 + 9 = 0$

$$\therefore \quad D^2 = -9 \Rightarrow D = \pm 3i$$

$$\therefore \quad y_C = C_1\cos 3x + C_2\sin 3x$$

Let $\quad y_P = C_1\cos 3x + C_2\sin 3x$

Here, cos 3x and sin 3x appears in y_C

$$y_P = x(C_1\cos 3x + C_2\sin 3x)$$

$\therefore$

$$y_P' = (C_1\cos 3x + C_2\sin 3x) + x(-3C_1\sin 3x + 3C_2\cos 3x)$$

$$y_P'' = (-3C_1\sin 3x + 3C_2\cos 3x) + (-3C_1\sin 3x + 3C_1\cos 3x) + x(-9C_1\cos 3x - 9C_2\sin 3x)$$

$\therefore$

Put y_P, y_P' and y_P'' in given equation

$$y_P'' + 9y_P = \mathbf{2\sin 3x}$$

Ex. 3.4.31 : *Solve $(D^2+3D+2)y = \sin 2x$.* **GTU : Winter-23, Maths-2, Marks 3**

Sol. : Here, $(D^2+3D+2)y = \sin 2x$ is given

A.E.

$$D^2 + 3D + 2 = 0$$

$$D^2 + 2D + D + 2 = 0$$

$$D(D+2) + 1(D+2) = 0$$

$$(D+1)(D+2) = 0$$

$$D = -1, -2$$

$$y_C = C_1e^{-x} + C_2e^{-2x}$$

Let us consider trail solution y_P

$\therefore$

$$y_P = A\cos 2x + B\sin 2x$$

$$y_P' = -2A\sin 2x + 2B\cos 2x$$

$\therefore$

$$y_P'' = -4A\cos 2x - 4B\sin 2x$$

Put y_P, y_P' and y_P'' in given equation

$\therefore$

$$(y_P'' + 3y_P' + 2y_P) = \sin 2x$$

$$-4A\cos 2x - 4B\sin 2x - 6A\sin 2x + 6B\cos 2x + 2A\cos 2x + 2B\sin 2x = \sin 2x$$

$$-2A\cos 2x - 2B\sin 2x - 6A\sin 2x + 6B\cos 2x = \sin 2x$$

$$\therefore (-2A+6B)\cos 2x + (-2B-6A)\sin 2x = \sin 2x$$

Comparing co-efficient

$$-2A + 6B = 0 \qquad -6A + 18B = 0$$

$$-6A - 2B = 1 \qquad -6A - 2B = 1$$

$$\qquad\qquad + \quad + \quad -$$

$$\therefore \quad -2A + 6\left(\frac{-1}{20}\right) = 0 \qquad 20B = -1$$

$$\therefore \quad \mathbf{B = \frac{-1}{20}}$$

$$\therefore \quad -2A = \frac{6}{20} \quad \Rightarrow \quad \mathbf{A = \frac{-3}{20}}$$

$$\therefore \quad y_P = \frac{-3}{20}\cos 2x - \frac{1}{20}\sin 2x$$

$$y = y_C + y_P$$

$$y = C_1 e^{-x} + C_2 e^{-2x} - \frac{3}{20}\cos 2x - \frac{1}{20}\sin 2x$$

Ex. 3.4.32 : *Using undetermined co-efficient method solve* $(D^2 + 4D)y = x + x^2$. **GTU : Winter-23, Maths-2, Marks 3**

Sol. : Here, given $(D^2 + 4D)y = x + x^2$

A.E. $D^2 + 4D = 0$

$$D(D+4) = 0$$

$$D = 0, D = -4$$

$$y_C = \text{C.F.} = C_1 + C_2 e^{-4x}$$

Let us consider trail solution y_P.

$$\therefore \quad y_P = Ax^2 + Bx + C$$

Here, constant term appear in y_C.

$$\therefore \quad y_P = Ax^3 + Bx^2 + Cx$$

$$y_P' = 3Ax^2 + 2Bx + C$$

$$\therefore \quad y_P'' = 6Ax + 2B$$

$\therefore$ y_P, y_P' and y_P'' in given equation,

$$\therefore \quad y_P'' + 4y_P' = x + x^2$$

$$6Ax + 2B + 12Ax^2 + 8Bx + 4C = x + x^2$$

$$\therefore \quad 12Ax^2 + (6A + 8B)x + (2B + 4C) = x + x^2$$

$$\therefore \quad 12A = 1 \Rightarrow \mathbf{A = \frac{1}{12}}$$

$$6A + 8B = 1 \Rightarrow 6\left(\frac{1}{12}\right) + 8B = 1 \qquad \Rightarrow \frac{1}{2} + 8B = 1$$

$$\therefore \quad 8B = 1 - \frac{1}{2} \Rightarrow 8B = \frac{1}{2} \Rightarrow \mathbf{B = \frac{1}{16}}$$

$$2B + 4C = 0 \Rightarrow 2\left(\frac{1}{16}\right) + 4C = 0 \qquad \Rightarrow \frac{1}{8} + 4C = 0$$

$$\Rightarrow 4C = \frac{-1}{8} \qquad \Rightarrow \mathbf{C = \frac{-1}{32}}$$

$$\therefore \quad y_P = \frac{1}{12}x^3 + \frac{1}{16}x^2 - \frac{1}{32}x$$

$$\therefore \quad y = y_C + y_P$$

$$y = C_1 + C_2 e^{-4x} + \frac{1}{12}x^3 + \frac{1}{16}x^2 - \frac{1}{32}x$$

Ex. 3.4.33 : *Solve* $y'' + y' - 12y = e^{6x}$. **GTU : Winter-21, Maths-2, Marks 4**

Sol. : Here, $y'' + y' - 12y = e^{6x}$ is given

A.E. $$(D^2 + D - 12) = 0$$

$$D^2 + 4D - 3D - 12 = 0$$

$$D(D+4) - 3(D+4) = 0$$

$$D = 3,\ D = -4$$

$$y_C = C_1 e^{3x} + C_2 e^{-4x}$$

Let us consider trail solution y_P

$$\therefore \quad y_P = C \cdot e^{6x},\ y_P' = 6Ce^{6x}$$

$$y_P'' = 36Ce^{6x}$$

Put y_P, y_P' and y_P'' in given equation

$$\therefore \quad y_P'' + y_P' - 12y_P = e^{6x}$$

$$36Ce^{6x} + 6Ce^{6x} - 12Ce^{6x} = e^{6x}$$

$$30Ce^{6x} = e^{6x}$$

$$\therefore \quad 30C = 1 \Rightarrow C = \frac{1}{30}$$

$$\therefore \quad y_P = \frac{1}{30} e^{6x}$$

$$\therefore \quad \mathbf{y = C_1 e^{3x} + C_2 e^{-4x} + \frac{1}{30} e^{6x}}$$

Ex. 3.4.34 : *Solve* $y'' - 4y' - 12y = \sin x$ *by method of undetermined co-efficient.* **GTU : Summer-20, AEM, Marks 4**

Sol. : Here, $y'' - 4y' - 12y = \sin x$ is given,

A.E. $$D^2 - 4D - 12 = 0$$

$$D^2 - 6D + 2D - 12 = 0$$

$$\therefore \quad D(D-6) + 2(D-6) = 0$$

$$(D+2)(D-6) = 0$$

$$\therefore \quad D = -2 \text{ and } D = 6$$

$$\therefore \quad y_C = C_1 e^{-2x} + C_2 e^{6x}$$

Now, let us consider trail solution.

$$y_P = A\cos x + B\sin x$$

$$y_P' = -A\sin x + B\cos x$$

$$y_P'' = -A\cos x - B\sin x$$

Put y_P, y'_P and y''_P in given equation.

$y''_P - 4y'_P - 12y_P = \sin x$

$-A\cos x - B\sin x + 4A\sin x - 4B\cos x - 12A\cos x - 12B\sin x = \sin x$

$-13A\cos x - 4B\cos x - 13B\sin x + 4A\sin x = \sin x$

$(-13A - 4B)\cos x + (4A - 13B)\sin x = \sin x$

Comparing co-efficient both sides,

$$\left.\begin{aligned} -13A - 4B &= 0 \\ 4A - 13B &= 1 \end{aligned}\right\} \quad \begin{aligned} -52A - 16B &= 0 \\ \underline{52A - 169B} &\underline{= 13} \\ -185B &= 13 \end{aligned}$$

$$\therefore \quad \mathbf{B} = -\frac{13}{185}$$

$$\therefore \quad 4A - 13\left(\frac{-13}{185}\right) = 1$$

$$\Rightarrow 4A = 1 - \frac{169}{185} \Rightarrow 4A = \frac{16}{185}$$

$$\Rightarrow \mathbf{A} = \frac{4}{185}$$

$$\therefore \quad y_P = \frac{4}{185}\cos x - \frac{13}{185}\sin x$$

$$\therefore \quad y = y_C + y_P = \mathbf{C_1 e^{-2x} + C_2 e^{6x} + \frac{4}{185}\cos x - \frac{13}{185}\sin x}$$

3.5 Method of Variation of Parameters

GTU : Summer-12,18,19,20,21,22,23,24, Winter-18,19,21,22,23, Marks 7

- **Method of variation of parameters :** Let us consider a differential equation of the form $f(D)\,y = r(x)$.
- When $r(x)$ is the form e^{ax}, sin ax, cos ax, x^m then we use undetermined co-efficient method for finding P.I. (y_P). If $r(x)$ be of any other form like tan x, sec x, cosec x cot x etc., then we have to use method of variation of parameters for finding y_P.

Consider second order linear ODE with constant co-efficient,

$$y'' + C_1 y' + C_2 y = r(x) \qquad \text{...(3.5.1)}$$

The C.F. (Complementary Function) of equation (3.5.1) is

$$\text{C.F.} = y_C = C_1 y_1 + C_2 y_2$$

Where, y_1 and y_2 will satisfy the equation

$$y'' + C_1 y' + C_2 y = 0$$

Now for finding P.I. (y_p) first find Wronskin's W.

If $W \neq 0$ then y_1 and y_2 are linearly independent.

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$W_1 = \begin{vmatrix} 0 & y_2 \\ 1 & y_2' \end{vmatrix} = -y_2$$

$$W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & 1 \end{vmatrix} = y_1$$

$$y_P = y_1 \int \frac{W_1}{W} r(x)\, dx + y_2 \int \frac{W_2}{W} r(x)\, dx$$

If differential equation of the form

$$\frac{d^3y}{dx^3} + C_1 \frac{d^2y}{dx^2} + C_2 \frac{dy}{dx} + C_3 = r(x) \qquad \text{...(3.5.2)}$$

The C.F. (Complimentary Function) of equation (3.5.2) is

$$\text{C.F.} = y_C = C_1y_1 + C_2y_2 + C_3y_3$$

$$\therefore \quad \text{Wronskin's } W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}.$$

$$W_1 = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ 1 & y_2'' & y_3'' \end{vmatrix}, \quad W_2 = \begin{vmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & 1 & y_3'' \end{vmatrix}$$

and $$W_3 = \begin{vmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & 1 \end{vmatrix}$$

$$\therefore \quad \mathbf{y_P = y_1 \int \frac{W_1}{W} r(x)\, dx + y_2 \int \frac{W_2}{W} r(x)\, dx + y_3 \int \frac{W_3}{W} r(x)\, dx}$$

Ex. 3.5.1 : *Using the method of variation of parameter solve the differential equation :* $(D^2 + 2D + 1)y = e^{-x} \cdot \log x$.

GTU : Summer-18, Maths-3, Marks 7

Sol. : Here, $(D^2 + 2D + 1)y = e^{-x} \cdot \log x$ is given

First, find C.F. (y_C) , here $r(x) = e^{-x} \cdot \log x$

$$\therefore \quad D^2 + 2D + 1 = 0$$

$$(D+1)^2 = 0$$

$$D = -1, -1 \text{ (}-1\text{ two times)}$$

$$\therefore \quad \text{C.F.} = (C_1 + C_2x)\, e^{-x}$$

$$\therefore \quad y_C = \text{C.F.} = C_1e^{-x} + C_2x \cdot xe^{-x}$$

Now, we know that the general form of C.F.

$$\text{C.F.} = C_1y_1 + C_2y_2$$

$\therefore \quad y_1 = e^{-x}$ and $y_2 = x \cdot e^{-x}$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{-x} & x e^{-x} \\ -e^{-x} & e^{-x} - x e^{-x} \end{vmatrix} = e^{-2x} - x e^{-2x} + x e^{-2x} = e^{-2x}$$

$$W_1 = -y_2 = -x \cdot e^{-x}$$

$$W_2 = y_1 = e^{-x}$$

Now, for particular integral the formula of variation of parameter is,

$$\text{P.I.} = y_P = y_1 \int \frac{W_1}{W} \cdot r(x)\, dx + y_2 \int \frac{W_2}{W} \cdot r(x)\, dx$$

$$\therefore \quad \text{P.I.} = y_P = e^{-x} \int \frac{-x \cdot e^{-x}}{e^{-2x}} \cdot e^{-x} \log x \, dx + x \cdot e^{-x} \int \frac{e^{-x}}{e^{-2x}} \cdot e^{-x} \log x \, dx$$

$$= -e^{-x} \int \underset{\substack{\uparrow \\ v}}{x} \cdot \underset{\substack{\uparrow \\ y}}{\log x} \, dx + x \cdot e^{-x} \int \log x \, dx$$

$\Rightarrow$ By integration of part we evaluate $\int x \cdot \log x \, dx$

$$= -e^{-x} \left[\log x \cdot \frac{x^2}{2} - \int \frac{1}{x} \cdot \frac{x^2}{2} dx \right] + x \cdot e^{-x} [x \log x - x] + C$$

$$= -e^{-x} \left[\log x \cdot \frac{x^2}{2} - \frac{x^2}{4} \right] + x \cdot e^{-x} [x \log x - x] + C$$

$$= -\frac{1}{2} \log x \cdot e^{-x} \cdot x^2 + \frac{1}{4} e^{-x} \cdot x^2 + x^2 e^{-x} \log x - x^2 e^{-x} + C$$

$$y_P = \frac{1}{2} x^2 e^{-x} \log x - \frac{3}{4} x^2 e^{-x} + C$$

$$\therefore \quad y = \text{C.F.} + \text{P.I.} = y_C + y_P$$

Ex. 3.5.2 : *Solve by method of variation of parameters, $y'' + a^2 y = \tan ax$.*

GTU : Summer-18, AEM, Marks 7, Summer-19, Maths-3, Marks 4, Summer-20,21 Maths-2, Marks 4, Winter-23, Maths-3, Marks 7, (a = 2)

Sol. : Here, $(D^2 + a^2)y = \tan ax$ is given

$\therefore$ The A.E. is $D^2 + a^2 = 0$

$$D^2 = -a^2 \Rightarrow D = \pm ia$$

$$\therefore \quad \text{C.F.} = y_P = C_1 \cos ax + C_2 \sin ax$$

$$\therefore \quad y_1 = \cos ax \text{ and } y_2 = \sin ax$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos ax & \sin ax \\ -a \sin ax & a \cos ax \end{vmatrix} = a \cos^2 ax + a \sin^2 ax = a \neq 0$$

$$W_1 = -y_2 = -\sin ax$$

$$W_2 = y_1 = \cos ax$$

Now, $$y_P = y_1\int \frac{W_1}{W}\cdot r(x)\,dx + y_2\int \frac{W_2}{W}\cdot r(x)\,dx$$

Where $r(x) = \tan ax$

$$\therefore \quad y_P = \cos ax\int \frac{-\sin ax}{a}\cdot \tan ax\,dx + \sin ax\int \frac{\cos ax}{a}\cdot \tan ax\,dx$$

$$= -\frac{\cos ax}{a}\int \frac{\sin^2 ax}{\cos ax}dx + \frac{\sin ax}{a}\int \sin ax\,dx$$

$$= -\frac{\cos ax}{a}\int \frac{1-\cos^2 ax}{\cos ax}dx + \frac{\sin ax}{a}\cdot\left(\frac{-\cos ax}{a}\right)$$

$$= -\frac{\cos ax}{a}\int \sec ax - \cos ax\,dx - \frac{\sin ax\cdot\cos ax}{a^2}$$

$$= -\frac{\cos ax}{a}\left[\frac{\log|\sec ax+\tan ax|}{a} - \frac{\sin ax}{a}\right] - \frac{\sin ax\cdot\cos ax}{a^2}$$

$$= \frac{-\cos ax}{a^2}\cdot\log|\sec ax+\tan ax| + \frac{\cos ax\cdot\sin ax}{a^2} - \frac{\sin ax\cdot\cos ax}{a^2}$$

$$\therefore \quad \text{P.I.} = y_P = -\frac{\cos ax}{a^2}\log|\sec ax + \text{tax}\ ax|$$

$$\therefore \quad y = \text{C.F.} + \text{P.I.}$$

$$y = \mathbf{C_1\cos ax + C_2\sin ax - \frac{\cos ax}{a^2}\log|\sec ax + \tan ax|}$$

Ex. 3.5.3 : *Using the method of variation of parameter, Solve $y'' - 4y' + 4y = \frac{e^{2x}}{x}$.*

GTU : Summer-20, Maths-3, Marks 4, Winter-22, AEM, Marks 7

Sol. : Here,

$$y'' - 4y' + 4y = \frac{e^{2x}}{x} \text{ is given}$$

$$\therefore \text{ A.E.} \quad D^2 - 4D + 4 = 0$$

$$\therefore \quad (D-2)^2 = 0$$

$$D = 2, 2$$

$$\therefore \quad \text{C.F.} = y_C = e^{2x}(C_1 + C_2 x)$$

$$y_C = C_1 e^{2x} + C_2 x e^{2x}$$

On comparing with C.F. $= C_1 y_1 + C_2 y_2$ we write $y_1 = e^{2x}$ and $y_2 = xe^{2x}$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{2x} & xe^{2x} \\ 2e^{2x} & e^{2x}+2xe^{2x} \end{vmatrix}$$

$$= e^{4x} + 2xe^{4x} - 2xe^{4x} = e^{4x} \neq 0$$

$$W_1 = -y_2 = -xe^{2x}$$

$$W_2 = y_1 = e^{2x}$$

$\therefore$ Method of V.P.

$$y_P = y_1 \int \frac{W_1}{W} \cdot r(x)\, dx + y_2 \int \frac{W_2}{W} \cdot r(x)\, dx$$

Where, $\quad r(x) = \dfrac{e^{2x}}{x}$

$$y_P = e^{2x} \cdot \int \frac{-x \cdot e^{2x}}{e^{4x}} \cdot \frac{e^{2x}}{x} dx + x e^{2x} \int \frac{e^{2x}}{e^{4x}} \cdot \frac{e^{2x}}{x} dx$$

$$= -e^{2x} \int 1\, dx + x e^{2x} \int \frac{1}{x} dx$$

$$\therefore \; y_P = -e^{2x} \cdot x + x \cdot e^{2x} \log x$$

$$\therefore \; y = y_C + y_P \text{ (Final Answer)}$$

$$\therefore \; y = \mathbf{e^{2x}(C_1 + C_2 x) + x\, e^{2x} \log x - x\, e^{2x}}$$

Ex. 3.5.4 : *Solve by method of variation of parameters,* $\dfrac{d^2 y}{dx^2} + y = \sec x.$

GTU : Summer-21, Maths-3, Marks 3.5, Summer-20, AEM, Marks 7

Sol. : Here, $y'' + y = \sec x$ is given

A.E. $\quad D^2 + 1 = 0$

$$D^2 = -1$$

$$\Rightarrow D = \pm i$$

$$\therefore \; \text{C.F.} = C_1 \cos x + C_2 \sin x$$

$$\therefore \; y_1 = \cos x \text{ and } y_2 = \sin x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$$

$$= \cos^2 x + \sin^2 x = 1 \neq 0$$

Now $\quad W_1 = -y_2 = -\sin x$

$$W_2 = y_1 = \cos x$$

$$\therefore \; y_P = y_1 \int \frac{W_1}{W} \cdot r(x)\, dx + y_2 \int \frac{W_2}{W} \cdot r(x)\, dx$$

Where, $r(x) = \sec x$

$$y_P = \cos x \int \frac{-\sin x}{1} \cdot \sec x\, dx + \sin x \int \frac{\cos x}{1} \cdot \sec x\, dx$$

$$\therefore \; y_P = -\cos x \log|\sec x| + \sin x \cdot x$$

$$\therefore \; y = y_C + y_P$$

$$= \mathbf{C_1 \cos x + C_2 \sin x - \cos x \cdot \log|\sec x| + x \cdot \sin x}$$

Ex. 3.5.5 : *Using the method of variation of parameters solve* $y'' - 3y' + 2y = e^x$.

GTU : Summer-20, AEM, Marks 3, Summer-22, Maths 3, Marks 4, Summer-24, Maths 3, Marks 4

Sol. : Here, $y'' - 3y' + 2y = e^x$ is given

A.E. $D^2 - 3D + 2 = 0$

$$(D-2)(D-1) = 0$$

$$\therefore \; D = 2 \text{ and } D = 1$$

$$\therefore \; y_C = \text{C.F.} = C_1 e^x + C_2 e^{2x}$$

Comparing C.F. $= C_1 y_1 + C_2 y_2$

$$\therefore \; y_1 = e^x \text{ and } y_2 = e^{2x}$$

$$W = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} \begin{matrix} = 2e^{3x} - e^{3x} \\ = e^{3x} \neq 0 \end{matrix}$$

$$W_1 = -y_2 = -e^{2x}$$

$$W_2 = y_1 = e^{2x}$$

Now, $\quad y_1 = y_1 \int \dfrac{W_1}{W} \cdot r(x)\, dx + y_2 \int \dfrac{W_2}{W} \cdot r(x)\, dx$

Where, $\quad r(x) = e^x$

$$\therefore \; y_1 = e^x \int \frac{-e^{2x}}{e^{3x}} e^x\, dx + e^{2x} \int \frac{ex}{e^{3x}} \cdot e^x dx$$

$$\therefore \; y_P = -e^x \int 1\, dx + e^{2x} \int e^{-x} dx$$

$$= -e^x \cdot x + e^{2x} \left(\frac{e^{-x}}{-1} \right) = -e^x \cdot x - e^x$$

$$= -e^x(x+1)$$

$$\therefore \; y = y_C + y_P = \mathbf{C_1 e^x + C_2 e^{2x} - e^x(x+1)}$$

Ex. 3.5.6 : *Using the method of variation of parameter, solve* $y'' + 9y = \sec 3x$.

GTU : Summer-19, Winter-21, Maths-2, Marks 7 [9 replace with 4], Summer-23, Maths-3, Marks 7, Summer-23, AEM, Marks 4

Sol. : Here, $y'' + 9y = \sec 3x$ is given

A.E. $\quad D^2 + 9 = 0$

$$D^2 = -9 \Rightarrow D = \pm 3i$$

$$y_C = C_1 \cos 3x + C_2 \sin 3x$$

Take, $y_1 = \cos 3x$ and $y_2 = \sin 3x$

$$W = \begin{vmatrix} \cos 3x & \sin 3x \\ -3\sin 3x & 3\cos 3x \end{vmatrix} \begin{matrix} = 3\cos^2 3x + 3\sin^2 3x \\ = 3 \neq 0 \end{matrix}$$

$$W_1 = -y_2 = -\sin 3x$$

$$W_2 = y_1 = \cos 3x$$

Using V.P. method, find y_P

$\therefore$ $$y_P = y_1 \int \frac{W_1}{W} \cdot r(x)\,dx + y_2 \int \frac{W_2}{W} \cdot r(x)\,dx$$

Where, $$r(x) = \sec 3x$$

$$y_P = \cos 3x \int \frac{-\sin 3x}{3} \cdot \sec 3x\,dx + \sin 3x \int \frac{\cos 3x}{3} \cdot \sec 3x\,dx$$

$$= -\frac{\cos 3x}{3} \frac{\log|\sec 3x|}{3} + \frac{\sin 3x}{3} \cdot x$$

$\therefore$ $$y = y_C + y_P = \mathbf{C_1 \cos 3x + C_2 \sin 3x - \frac{\cos 3x}{9} \log|\sec 3x| + \frac{\sin 3x}{3} \cdot x}$$

Ex. 3.5.7 : *Solve* $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = \frac{e^x}{1+e^x}$ *using method of variation of parameter.*

GTU : Winter-18, Maths-3, Marks 4, Winter-21, Maths-3, Marks 7

Sol. : Here,

$$y'' - 3y' + 2y = \frac{e^x}{1+e^x} \text{ is given}$$

A.E. $$D^2 - 3D + 2 = 0$$

$$(D-2)(D-1) = 0$$

$$D = 1 \text{ and } D = 2$$

$\therefore$ $$\text{C.F.} = y_C = C_1 e^x + C_2 e^{2x}$$

Comparing with

$$\text{C.F.} = C_1 y_1 + C_2 y_2$$

$\therefore$ $$y_1 = e^x \text{ and } y_2 = e^{2x}$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$\therefore$ $$W = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} \begin{matrix} = 2e^{3x} - e^{3x} \\ = e^{3x} \neq 0 \end{matrix}$$

$$\therefore \quad W_1 = -y_2 = -e^{2x}$$

$$W_2 = y_1 = e^x$$

Variation of parameter is used for finding y_P

$$\therefore \quad y_P = y_1 \int \frac{W_1}{W} \cdot r(x)\, dx + y_2 \int \frac{W_2}{W} \cdot r(x)\, dx$$

Where, $$r(x) = \frac{e^x}{1+e^x}$$

$$y_P = e^x \int \frac{-e^{2x}}{e^{3x}} \cdot \frac{e^x}{1+e^x} dx + e^{2x} \int \frac{e^x}{e^{3x}} \cdot \frac{e^x}{1+e^x} dx$$

$$= -e^x \int \frac{1}{1+e^x} dx + e^{2x} \int \frac{1}{e^x(1+e^x)} dx$$

$$= -e^x \int \frac{e^{-x}}{1+e^{-x}} dx + e^{2x} \int \left(e^{-x} - \frac{e^{-x}}{1+e^{-x}} \right) dx$$

$$= -e^x(-\log(1+e^{-x})) + e^{2x}(-e^{-x} + \log(1+e^{-x}))$$

$$= e^x \log(1+e^{-x}) - e^x + e^{2x} \log(1+e^{-x})$$

$$\therefore \quad y = y_C + y_P$$

$$\mathbf{y = C_1 e^x + C_2 e^{2x} + e^x \log(1+e^{-x}) - e^x + e^{2x} \log(1+e^{-x})}$$

Ex. 3.5.8 : *Solve by method of variation of parameter* $y'' + 4y = \sec 2x$.

GTU : Winter-19, Maths-3, Marks 7

Sol. : Here, $y'' + 4y = \sec 2x$ is given

A.E. $$D^2 + 4 = 0 \Rightarrow D = \pm 2i$$

$$\therefore \quad y_C = C_1 \cos 2x + C_2 \sin 2x$$

$$\therefore \quad y_1 = \cos 2x \qquad y_1 = \sin 2x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos 2x & \sin 2x \\ -2\sin 2x & 2\cos 2x \end{vmatrix}$$

$$= 2\cos^2 2x + 2\sin^2 2x = 2 \neq 0$$

$$W_1 = -y_2 = -\sin 2x$$

$$W_2 = y_1 = \cos 2x$$

$$\therefore \quad y_P = y_1 \int \frac{W_1}{W} \cdot r(x)\, dx + y_2 \int \frac{W_2}{W} \cdot r(x)\, dx$$

Where, $$r(x) = \sec 2x$$

$$\therefore \quad y_P = \cos 2x \int \frac{-\sin 2x}{2} \cdot \sec 2x\, dx + \sin 2x \int \frac{\cos 2x}{2} \cdot \sec 2x\, dx$$

$$\therefore \quad y_P = -\frac{\cos 2x}{2}\frac{\log|\sec 2x|}{2}+\frac{\sin 2x}{2}\cdot x$$

$$\therefore \quad y = y_C + y_P$$

$$= C_1\cos 2x + C_2\sin 2x - \frac{\cos 2x}{4}\log|\sec 2x| + \frac{x\cdot\sin 2x}{2}$$

Ex. 3.5.9 : *Using the method of variation of parameters find the general solution of the differential equation $y''+y = \sin x$.*

GTU : Winter-22, Maths-3, Summer-24, Maths-2, Marks 7

Sol. : Here, $y''+4y = \sin x$ is given,

A.E. $$D^2+1 = 0$$

$$\Rightarrow D^2 = -1 \Rightarrow D = \pm i$$

$$\therefore \quad y_C = C_1\cos x + C_2\sin x$$

Now, we use variation of parameter to find y_P

$$\therefore \quad \text{Let } y_1 = \cos x \text{ and } y_2 = \sin x$$

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} \begin{matrix} = \cos^2 x + \sin^2 x \\ = 1 \neq 0 \end{matrix}$$

$$W_1 = -y_2 = -\sin x \text{ and } W_2 = y_1 = \cos x$$

$$\therefore \quad y_P = y_1\int\frac{W_1}{W}\cdot r(x)\,dx + y_2\int\frac{W_2}{W}\cdot r(x)\,dx$$

Where, $r(x) = \sin x$

$$\therefore \quad y_P = \cos x\int\frac{-\sin x}{1}\cdot\sin x\,dx + \sin x\int\frac{\cos x}{1}\cdot\sin x\,dx$$

$$= -\cos x\int\sin^2 x\,dx + \frac{\sin x}{2}\int\sin 2x\,dx$$

$$= -\cos x\int\frac{1-\cos 2x}{2}dx + \frac{\sin x}{2}\left(\frac{-\cos 2x}{2}\right)$$

$$= -\frac{\cos x}{2}\left(x-\frac{\sin 2x}{2}\right) - \frac{\sin x\cdot\cos 2x}{4}$$

$$y_P = -\frac{\cos x\cdot x}{2} + \frac{\cos x\cdot\sin 2x}{4} - \frac{\sin x\cdot\cos 2x}{4}$$

Ex. 3.5.10 : *Solve by method of variation of parameters $\left(\frac{d^2y}{dx^2}+9y\right) = \frac{1}{1+\sin 3x}$.*

GTU : Summer-19, AEM, Marks 7

Sol. : Here,

$$y''+9y = \frac{1}{1+\sin 3x} \text{ is given}$$

A.E. $$D^2+9 = 0$$

$$\therefore \quad D^2 = -9 \Rightarrow D = \pm 3i$$

$$y_C = C_1\cos 3x + C_2\sin 3x$$

Comparing with C.F. $= C_1y_1 + C_2y_2$

$$\therefore \quad y_1 = \cos 3x \quad y_2 = \sin 3x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos 3x & \sin 3x \\ -3\sin 3x & 3\cos 3x \end{vmatrix} = 3\cos^2 3x + 3\sin^2 3x = 3 \neq 0$$

$$W_1 = -y_2 = -\sin 3x$$

$$W_2 = y_1 = \cos 3x$$

$$\therefore \quad y_P = y_1\int \frac{W_1}{W}\cdot r(x)\,dx + y_2\int \frac{W_2}{W}\cdot r(x)\,dx$$

Where, $\quad r(x) = \dfrac{1}{1+\sin 3x}$

$$y_P = \cos 3x\int \frac{-\sin 3x}{3}\cdot\frac{1}{1+\sin 3x}\,dx + \sin 3x\int \frac{\cos 3x}{3}\cdot\frac{1}{1+\sin 3x}\,dx$$

$$y_P = -\frac{\cos 3x}{3}\int \frac{\sin 3x}{1+\sin 3x}\,dx + \frac{\sin 3x}{3}\int \frac{\cos 3x}{1+\sin 3x}\,dx$$

$$\therefore \quad y_P = \frac{-\cos 3x}{3}\int 1 - \frac{1}{1+\sin 3x}\,dx + \frac{\sin 3x}{9}\int \frac{3\cdot\cos 3x}{1+\sin 3x}\,dx \quad \left(\because \int \frac{f'(x)}{f(x)}\,dx = \log f(x)\right)$$

$$y_P = -\frac{\cos 3x}{3}\left(x - \frac{2}{3+3\tan\left(\frac{3x}{2}\right)}\right) + \frac{\sin 3x}{9}(\log(1+\sin 3x))$$

$$\therefore \quad y = y_C + y_P = \mathbf{C_1\cos 3x + C_2\sin 3x - \frac{\cos 3x}{3}\cdot x + \frac{2}{9}\cdot\frac{\cos 3x}{1+\tan\left(\frac{3x}{2}\right)} + \frac{\sin 3x}{9}(\log(1+\sin 3x))}$$

Ex. 3.5.11 : *Using the method of variation of parameter find the general solution of* $(D^2 - 2D + 1)y = 3x^{3/2}e^x$.

GTU : Winter-19, Summer-24, AEM, Marks 7

Sol. : Here, $(D^2 - 2D + 1)y = 3x^{3/2}e^x$ is given

$$\therefore \text{ A.E.} \quad D^2 - 2D + 1 = 0$$

$$(D-1)^2 = 0$$

$$\therefore \quad D = 1, 1$$

$$\therefore \quad y_C = e^x(C_1 + C_2x) = C_1e^x + C_2xe^x$$

Now, $\quad y_1 = e^x \quad y_2 = xe^x$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & xe^x \\ e^x & e^x + xe^x \end{vmatrix} = e^{2x} + xe^{2x} - xe^{2x} = e^{2x} \neq 0$$

$$W_1 = -y_2 = -xe^x$$

$$W_2 = y_1 = e^x$$

$\therefore \quad y_P = y_1\int \frac{W_1}{W}\cdot r(x)\,dx + y_2\int \frac{W_2}{W}\cdot r(x)\,dx$

Where, $\quad r(x) = 3x^{3/2}e^x$

$\therefore \quad y_P = e^x\int \frac{-xe^x}{e^{2x}}\cdot 3x^{3/2}e^x dx + xe^x\int \frac{e^x}{e^{2x}}\cdot 3x^{3/2}e^x dx$

$\therefore \quad y_P = -3e^x\int x^{5/2}dx + 3xe^x\int x^{3/2}dx = -3e^x\left[\frac{x^{7/2}}{7/2}\right] + 3xe^x\left[\frac{x^{5/2}}{5/2}\right]$

$= -\frac{6}{7}e^x x^{7/2} + \frac{6}{5}e^x x^{7/2} = \frac{12}{35}e^x \cdot x^{7/2}$

$\therefore \quad y = y_C + y_P$

$\therefore \quad \mathbf{y = e^x(C_1 + C_2x) + \frac{12}{35}e^x \cdot x^{7/2}}$

Ex. 3.5.12 : *Solve using - method of variation of parameters $y'' + 2y' + y = e^{-x}\cos x$.* **GTU : Winter-18, AEM, Marks 7**

Sol. : Here, $y'' + 2y' + y = e^{-x}\cos x$ is given

A.E. $\quad D^2 + 2D + 1 = 0$

$\therefore \quad (D+1)^2 = 0$

$\therefore \quad D = -1, -1$

$\therefore \quad y_C = e^{-x}(C_1 + C_2x)$

Comparing this with C.F. $= C_1y_1 + C_2y_2$

$\therefore \quad y_1 = e^{-x}$ and $y_2 = xe^{-x}$

$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{-x} & xe^{-x} \\ -e^{-x} & e^{-x} - xe^{-x} \end{vmatrix}$

$= e^{-2x} - xe^{-2x} + xe^{-2x} = e^{-2x} \neq 0$

$W_1 = -y_2 = -xe^{-x}$

$W_2 = y_1 = e^{-x}$

$\therefore \quad y_P = y_1\int \frac{W_1}{W}\cdot r(x)\,dx + y_2\int \frac{W_2}{W}\,r(x)\,dx$

where, $\quad r(x) = e^{-x}\cos x$

$\therefore \quad y_P = e^{-x}\int \frac{-xe^{-x}}{e^{-2x}}\cdot e^{-x}\cos x\,dx + xe^{-x}\int \frac{e^{-x}}{e^{-2x}}e^{-x}\cos x\,dx$

$\therefore \quad y_P = -e^{-x}\int x\cos x\,dx + xe^{-x}\int \cos x\,dx = -e^{-x}[x\sin x - (-\cos x)] + xe^{-x}(\sin x)$

$= -xe^{-x}\sin x - e^{-x}\cos x + xe^{-x}\sin x$

$\therefore \quad y_P = -e^{-x}\cos x$

$\therefore \quad y = y_C + y_P = \mathbf{e^{-x}(C_1 + C_2x) - e^{-x}\cos x}$

Ex. 3.5.13 : *Solve the differential equation $y''' + y' = \text{cosec } x$ by the method of variation of parameter.*

GTU : Summer-12, Maths-3, Marks 7

Sol. : Here, $y''' + y' = \text{cosec } x$ is given

A.E. $$D^3 + D = 0$$

$$D(D^2+1) = 0$$

$$D = 0 \text{ and } D^2 + 1 = 0$$

$$D^2 = -1$$

$\therefore$ $$D = \pm i$$

$\therefore$ $$\text{C.F.} = y_C = C_1 + C_2 \cos x + C_3 \sin x$$

Comparing with $y_C = C_1 y_1 + C_2 y_2 + C_3 y_3$

$\therefore$ $$y_1 = 1, \quad y_2 = \cos x, \quad y_3 = \sin x$$

$\therefore$ $$W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 1 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ 0 & -\cos x & -\sin x \end{vmatrix}$$

$$= \sin^2 x + \cos^2 x = 1 \neq 0$$

Now, $$W_1 = \begin{vmatrix} 0 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ 1 & -\cos x & -\sin x \end{vmatrix} = \sin^2 x + \cos^2 x = 1 \neq 0$$

$$W_2 = \begin{vmatrix} 1 & 0 & \sin x \\ 0 & 0 & \cos x \\ 0 & 1 & -\sin x \end{vmatrix} = -\cos x$$

$$W_3 = \begin{vmatrix} 1 & \cos x & 0 \\ 0 & -\sin x & 0 \\ 0 & -\cos x & 1 \end{vmatrix} = -\sin x$$

$$y_P = y_1 \int \frac{W_1}{W} r(x)\,dx + y_2 \int \frac{W_2}{W} r(x)\,dx + y_3 \int \frac{W_3}{W} r(x)\,dx$$

$$= 1 \int \frac{1}{1} \cdot \text{cosec } x\,dx + \cos x \int \frac{-\cos x}{1} \cdot \text{cosec } x\,dx + \sin x \int \frac{-\sin x}{1} \cdot \text{cosec } x\,dx$$

$$= \log|\text{cosec } x - \cot x - \cos x \log|\sin x| - \sin x \cdot x$$

$$y_P = C_1 + C_2 \cos x + C_3 \sin x + \log|\text{cosec } x - \cot x| - \cos x \log|\sin x| - x \cdot \sin x$$

$$y = y_C + y_P$$

$$= \mathbf{C_1 + C_2 \cos x + C_3 \sin x + \log|\text{cosec } x - \cot x| - \cos x \log|\sin x| - x\sin x}$$

Ex. 3.5.14 : *Apply the method of variation of parameter to solve*
$y''' - 9y'' + 27y' - 27y = \dfrac{e^{3x}}{x^2}$.

Sol. :

Here, $$y''' - 9y'' + 27y' - 27y = \frac{e^{3x}}{x^2}$$

A.E. $$D^3 - 9D^2 + 27D - 27 = 0$$

$$(D-3)^3 = 0$$

$$D = 3, 3, 3$$

$$\text{C.F.} = y_C = e^{3x}(C_1 + C_2 x + C_3 x^2)$$

$$\text{C.F.} = C_1 e^{3x} + C_2 x e^{3x} + C_3 x^2 e^{3x}$$

$$\text{Comparing with C.F.} = C_1 y_1 + C_2 y_2 + C_3 y_3$$

$$\therefore \quad y_1 = e^{3x},\ y_2 = x \cdot e^{3x},\ y_3 = x^2 e^{3x}$$

$$\therefore \quad W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} e^{3x} & xe^{3x} & x^2 e^{3x} \\ 3e^{3x} & e^{3x} + 3xe^{3x} & (3x^2+2)e^{3x}x \\ 9e^{3x} & 9xe^{3x} + 6e^{3x} & (9x^2+12x+2)e^{3x} \end{vmatrix}$$

$$= e^{9x} \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & 2x \\ 0 & 6 & 12x+2 \end{vmatrix} = 2 \cdot e^{9x}$$

$$W_1 = \begin{vmatrix} 0 & xe^{3x} & x^2 e^{3x} \\ 0 & e^{3x} + 3xe^{3x} & (3x^2+2)e^{3x} \cdot x \\ 1 & 9xe^{3x} + 6e^{3x} & (9x^2+12x+2)e^{3x} \end{vmatrix} = x^2 e^{6x}$$

$$W_2 = \begin{vmatrix} e^{3x} & 0 & x^2 e^{3x} \\ 3e^{3x} & 0 & (3x^2+2)e^{3x}x \\ 9e^{3x} & 1 & (9x^2+12x+2)e^{3x} \end{vmatrix} = -2xe^{6x}$$

$$W_3 = \begin{vmatrix} e^{3x} & xe^{3x} & 0 \\ 3e^{3x} & e^{3x} + 3xe^{3x} & 0 \\ 9e^{3x} & 9xe^{3x} + 6e^{3x} & 1 \end{vmatrix} = e^{6x}$$

$$\therefore \quad y_P = y_1 \int \frac{W_1}{W} r(x)\, dx + y_2 \int \frac{W_2}{W} r(x)\, dx + y_3 \int \frac{W_3}{W} r(x)\, dx$$

$$= e^{3x} \int \frac{x^2 e^{6x}}{2e^{9x}} \cdot \frac{e^{3x}}{x^2}\, dx + x \cdot e^{3x} \int \frac{-2x e^{6x}}{2e^{9x}} \cdot \frac{e^{3x}}{x^2}\, dx + x^2 e^{3x} \int \frac{e^{6x}}{2e^{9x}} \cdot \frac{e^{3x}}{x^2}\, dx$$

$$= \frac{e^{3x}}{2} \int 1 dx + x \cdot e^{3x} \int \frac{-1}{x}\, dx + \frac{x^2 e^{3x}}{2} \int \frac{1}{x^2}\, dx$$

$$y_P = \frac{e^{3x}}{2} x - xe^{3x} \log x - \frac{x^2 e^{3x}}{2}\left(-\frac{1}{x}\right)$$

$$\therefore \quad y_P = -xe^{3x} \log x$$

$$\therefore \quad y = y_C + y_P = C_1 e^{3x} + C_2 x e^{3x} + C_3 x^2 e^{3x} - x e^{3x} \log x$$

Ex. 3.5.15 : *Solve $y'' + y = \operatorname{cosec} x$ using variation of parameter method.* **GTU : Winter-21, AEM, Marks 7**

Sol. : Here, $y'' + y = \operatorname{cosec} x$ is given

A.E. $$D^2 + 1 = 0$$

$$D^2 = -1$$

$$\therefore \quad D = \pm i$$

$$\therefore \quad y_C = C_1 \cos x + C_2 \sin x$$

Compare with C.F. $= C_1 y_1 + C_2 y_2$

$$\therefore \quad y_1 = \cos x \quad y_2 = \sin x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1 \neq 0$$

$$W_1 = -y_2 = -\sin x$$

$$W_2 = y_1 = \cos x$$

$$y_P = y_1 \int \frac{W_1}{W} r(x)\, dx + y_2 \int \frac{W_2}{W} r(x)\, dx$$

where, $$r(x) = \operatorname{cosec} x$$

$$\therefore \quad y_P = \cos x \int \frac{\sin x}{1} \cdot \operatorname{cosec} x\, dx + \sin x \int \frac{\cos x}{1} \cdot \operatorname{cosec} x\, dx$$

$$= -\cos x \int 1\, dx + \sin x \int \cot x\, dx$$

$$= -\cos x \cdot x + \sin x \log|\sin x|$$

$$y = y_C + y_P$$

$$\mathbf{y = C_1 \cos x + C_2 \sin x - x \cdot \cos x + \sin x \log|\sin x|}$$

Ex. 3.5.16 : *Using method of variation of parameter solve $y'' - 2y' + y = xe^x \sin x$.* **GTU : Winter-23, Maths-2, Marks 7**

Sol. : Here, $y'' - 2y' + y = xe^x \sin x$ is given

A.E. $$D^2 - 2D + 1 = 0$$

$$(D-1)^2 = 0$$

$$D = 1, 1$$

$$\therefore \quad y_C = e^x (C_1 + C_2 x)$$

$$\therefore \quad y_C = C_1 e^x + C_2 x e^x$$

Let, $$y_1 = e^x \quad y_2 = x e^x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & xe^x \\ e^x & e^x + xe^x \end{vmatrix}$$

$$= e^{2x} + xe^{2x} - xe^{2x} = e^{2x} \neq 0$$

$$W_1 = -y_2 = -xe^x$$

$$W_2 = y_1 = e_x$$

$$y_P = y_1 \int \frac{W_1}{W} \cdot r(x)\,dx + y_2 \int \frac{W_2}{W} \cdot r(x)\,dx$$

$$y_P = e^x \int \frac{-xe^x}{e^{2x}} \cdot xe^x \sin x\,dx + xe^x \int \frac{e^x}{e^{2x}} xe^x \sin x\,dx$$

$$= -e^x \int x^2 \sin x\,dx + xe^x \int x \sin x\,dx$$

$$= -e^x[(x^2(-\cos x) - (2x)(-\sin x) + (2)(\cos x)] + xe^x[(x)(-\cos x) - (1)(-\sin x)]$$

$$= e^x x^2 \cos x - 2e^x x \sin x - 2e^x \cos x - x^2 e^x \cos x + xe^x \sin x$$

$$= -e^x x \sin x - 2e^x \cos x$$

$$= -e^x(x \sin x + 2\cos x)$$

$$y = y_C + y_P$$

$$\mathbf{y = C_1 e^x + C_2 x e^x - e^x(x \sin x + 2\cos x)}$$

Ex. 3.5.17 : *Solve the differential equation : $(D^2 - 2D + 1)y = e^x$ using method of variation of parameters.*

Sol. : Auxiliary equation is : $(m^2 - 2m + 1) = 0$

$$\Rightarrow \quad m = 1, 1$$

$$\text{C.F.} = (C_1 + C_2 x)e^x = C_1 e^x + C_2 x e^x = C_1 y_1 + C_2 y_2$$

$$\therefore \quad y_1 = e^x \text{ and } y_2 = xe^x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & xe^x \\ e^x & xe^x + e^x \end{vmatrix} = e^{2x}$$

$$u_1 = -\int \frac{y_2 F(x)}{W} dx = -\int \frac{xe^x e^x}{e^{2x}} dx$$

$$= -\int x\,dx = -\frac{x^2}{2}$$

$$u_2 = \int \frac{y_1 F(x)}{W} dx = \int \frac{e^x e^x}{e^{2x}} dx = \int 1\,dx = x$$

$$\therefore \quad \text{P.I.} = u_1 y_1 + u_2 y_2$$

$$= -\frac{x^2}{2} e^x + x^2 e^x = \frac{x^2}{2} e^x$$

Complete solution is : y = C.F. + P.I.

$$\Rightarrow \quad \mathbf{y = (C_1 + C_2 x)e^x + \frac{x^2}{2} e^x}$$

Ex. 3.5.18 : *Solve the differential equation :*
$\dfrac{d^2y}{dx^2}+4y = x \sin 2x$ *using method of variation of parameters.*

Sol. :

$$\Rightarrow \qquad (D^2+4)\,y = x\sin 2x$$

Auxiliary equation is : $(m^2+4) = 0$

$$\Rightarrow \qquad m = \pm 2i$$

$$\text{C.F.} = C_1 \cos 2x + C_2 \sin 2x = C_1 y_1 + C_2 y_2$$

$$\therefore \qquad y_1 = \cos 2x \quad \text{and} \quad y_2 = \sin 2x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos 2x & \sin 2x \\ -2\sin 2x & 2\cos 2x \end{vmatrix} = 2$$

$$u_1 = -\int \frac{y_2 F(x)}{W}\,dx = -\frac{1}{2}\int x\sin^2 2x\,dx = -\frac{1}{4}\int x(1-\cos 4x)\,dx$$

$$= -\frac{1}{4}\left[\frac{x^2}{2}-\left[(x)\left(\frac{\sin 4x}{4}\right)-(1)\left(-\frac{\cos 4x}{16}\right)\right]\right] = \left[-\frac{x^2}{8}+\frac{x\sin 4x}{16}+\frac{\cos 4x}{64}\right]$$

$$u_2 = \int \frac{y_2 F(x)}{W}\,dx = \frac{1}{2}\int x\sin 2x\cos 2x\,dx = \frac{1}{4}\int x\sin 4x\,dx$$

$$= \frac{1}{4}\left[(x)\left(-\frac{\cos 4x}{4}\right)-(1)-\left(-\frac{\sin 4x}{16}\right)\right] = \left[-\frac{x\cos 4x}{16}+\frac{\sin 4x}{64}\right]$$

$$\therefore \qquad \text{P.I.} = u_1 y_1 + u_2 y_2$$

$$= \cos 2x\left[-\frac{x^2}{8}+\frac{x\sin 4x}{16}+\frac{\cos 4x}{64}\right]+\sin 2x\left[-\frac{x\cos 4x}{16}+\frac{\sin 4x}{64}\right]$$

$$= \frac{x}{16}(\sin 4x\cos 2x-\cos 4x\sin 2x)+\frac{1}{64}(\cos 4x\cos 2x+\sin 4x\sin 2x)-\frac{x^2}{8}\cos 2x$$

$$= \frac{x}{16}\sin 2x+\frac{1}{64}\cos 2x-\frac{x^2}{8}\cos 2x$$

Complete solution is : $y = \text{C.F.} + \text{P.I.}$

$$\Rightarrow \qquad \mathbf{y = C_1\cos 2x + C_2\sin 2x + \frac{x}{16}\sin 2x + \frac{1}{64}\cos 2x - \frac{x^2}{8}\cos 2x}$$

Ex. 3.5.19 : *Solve the differential equation :* $(D^2-D-2)y = e^{(e^x+3x)}$ *using method of variation of parameters.*

Sol. : Auxiliary equation is : $(m^2-m-2) = 0$

$$\Rightarrow \qquad m = -1, 2$$

$$\text{C.F.} = C_1 e^{-x} + C_2 e^{2x} = C_1 y_1 + C_2 y_2$$

$$\therefore \qquad y_1 = e^{-x} \quad \text{and} \quad y_2 = e^{2x}$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{-x} & e^{2x} \\ -e^{-x} & 2e^{2x} \end{vmatrix} = 3e^x$$

$$u_1 = -\int \frac{y_2 F(x)}{W} dx = -\int \frac{e^{2x} e(e^x + 3x)}{3e^x} dx = -\int \frac{e^{2x} e^{e^x} e^{3x}}{3e^x} dx$$

$$= -\frac{1}{3}\int e^{4x} e^{e^x} dx, \text{ Putting } e^x = t \Rightarrow e^x dx = t\,dt$$

$$u_1 = -\frac{1}{3}\int t^3 e^t dt = -\frac{1}{3}[(t^3)(e^t) - (3t^2)(e^t) + (6t)(e^t) - (6)(e^t)]$$

$$\Rightarrow \quad u_1 = -\frac{e^{e^x}}{3}[e^{3x} - 3e^{2x} + 6e^x - 6]$$

$$u_2 = \int \frac{y_1 F(x)}{W} dx = \int \frac{e^{-x} e^{(e^x + 3x)}}{3e^x} dx = \int \frac{e^{-x} e^{e^x} e^{3x}}{3e^x} dx \frac{1}{3}\int e^x e^{e^x} dx = \frac{e^{e^x}}{3}$$

$$\therefore \quad \text{P.I.} = u_1 y_1 + u_2 y_2$$

$$= -\frac{e^{e^x} e^{-x}}{3}[e^{3x} - 3e^{2x} + 6e^x - 6] + \frac{e^{e^x} e^{2x}}{3} = \frac{e^{e^x}}{3}[3e^x - 6 + 6e^{-x}]$$

Complete solution is : $y = \text{C.F.} + \text{P.I.}$

$$\Rightarrow \quad y = C_1 e^{-x} + C_2 e^{2x} + \frac{e^{e^x}}{3}[3e^x - 6 + 6e^{-x}]$$

3.6 Euler-Cauchy Equations

GTU : Winter-11,12,17,22,23, Summer-12,18,19,20,22,23,24

Euler Cauchy's Equations :

- A differential equation of the form

$$x^n \frac{dy^n}{dx^n} + C_1 x^{n-1} \frac{dy^{n-1}}{dx^{n-1}} + C_2 x^{n-2} \frac{dy^{n-2}}{dx^{n-2}} + \ldots + C_n y = f(x) \quad \ldots(3.6.1)$$

- Where, $C_1, C_2, \ldots, C_n$ are constant and $f(x)$ is a function of x is called **Euler Cauchy's** linear equation of order n.
- Now equation (3.6.1) can be convert in to linear differential equation with constant coefficient by substituting.

$$x = e^z \quad \text{or} \quad z = \log x$$

Now,
$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx}$$

$$z = \log x \Rightarrow \frac{dz}{dx} = \frac{1}{x}$$

$$\therefore \quad \frac{dy}{dx} = \frac{1}{x} \cdot \frac{dy}{dz}$$

$$\therefore \quad x \cdot \frac{dy}{dx} = \frac{dy}{dz} = Dy, \quad \text{where } D = \frac{d}{dz}$$

Similarly, $x^2 \dfrac{d^2y}{dx^2} = D(D-1)y$

$$x^3 \frac{d^3y}{dx^3} = D(D-1)(D-2/y) \text{ and so on}$$

- Substituting these value in equation (3.6.1), it can be convert to linear differential equation with constant co-efficient.
- This types of equation solved by the earlier discussed method.

Ex. 3.6.1 : *Solve* $(x^2 D^2 + 3xD + 4)y = 0$, $y(1) = 0$, $y'(1) = 3$.

GTU : Winter-11, Maths-3, Marks 4
Summer-22, Maths-2, Marks 4

Sol. : Here, $(x^2D^2 - 3xD + 4)\,y = 0$ is given

Let, $x = e^z \Rightarrow z = \log x$

$\therefore \quad x \cdot \dfrac{dy}{dx} = Dy$ and $x^2 \dfrac{d^2y}{dx^2} = D(D-1)$

Where, $D = \dfrac{d}{dz}$

$\therefore$ Given equation can be express as,

$$(D(D-1) - 3D + 4)y = 0$$

$$\therefore \quad (D^2 - 4D + 4)y = 0$$

$$\therefore \text{ A.E.} \quad D^2 - 4D + 4 = 0$$

$$(D-2)^2 = 0$$

$$\therefore \quad D = 2, 2$$

$$\text{C.F.} = y = e^{2z}(C_1 + C_2 z)$$

$$\therefore \quad y = x^2(C_1 + C_2 \log x)$$

Now, initial condition is given,

$y(1) = 0 \Rightarrow y = 0$ when $x = 1$

$$0 = 1(C_1 + C_2 \log 1)$$

$$\mathbf{C_1 = 0}$$

Take derivative of y.

$$y' = 2x(C_1 + C_2 \log x) + x^2\left(C_2 + \frac{1}{x}\right)$$

Here, $y'(1) = 3$ is given

$\Rightarrow \quad y' = 3$ when $x = 1$

$$\therefore \quad 3 = 2C_1 + C_2$$

$$\therefore \quad C_2 = 3$$

$$\therefore \quad \mathbf{y = 3x^2 \log x}$$

Ex. 3.6.2 : *Solve* $(4x^2 D^2 + 16xD + 9)y = 0$.

GTU : Winter-23, AEM, Marks 7

Sol. : Here, $(4x^2D^2 + 16xD + 9)\,y = 0$ is given

This is Euler Cauchy's equation.

$\therefore$ Let, $x = e^z \Rightarrow z = \log x$

$$x \cdot \frac{dy}{dx} = Dy$$

$$x^2 \cdot \frac{d^2y}{dx^2} = D(D-1)y \quad \text{where, } D = \frac{d}{dz}$$

$$\therefore \quad (4D(D-1) + 16D + 9)\,y = 0$$

$$\therefore \quad (4D^2 + 12D + 9)\,y = 0$$

$$\therefore \text{ A.E.} \quad 4D^2 + 12D + 9 = 0$$

$$4D^2 + 6D + 6D + 9 = 0$$

$$2D(2D+3) + 3(2D+3) = 0$$

$$\therefore \quad (2D+3)^2 = 0$$

$$\therefore \quad 2D = -3 \Rightarrow D = \frac{-3}{2}$$

$$\therefore \quad y_C = (C_1 + C_2 z)\,e^{-3/2^z}$$

$$\therefore \quad \mathbf{y = (C_1 + C_2 \log z)\,x^{-3/2}}$$

Ex. 3.6.3 : $x^2 y'' - 20y = 0$.

GTU : Summer-24, Mathe-2, Marks 3

Sol. : Here, $x^2y'' - 20y = 0$

This is a Euler's Cauchy equation,

$\therefore \quad x = e^z \Rightarrow z = \log x$

$$x\frac{dy}{dx} = Dy, \quad x^2 \frac{d^2y}{dx^2} = D(D-1)y$$

$$\therefore \quad (D(D-1) - 20)y = 0$$

$$\therefore \quad (D^2 - D - 20)\,y = 0$$

A.E. $D^2 - D - 20 = 0$

$$D^2 - 5D + 4D - 20 = 0$$

$$D(D-5) + 4(D-5) = 0$$

$$(D+4)(D-5) = 0$$

$$D = 4, 5$$

$$y = C_1 e^{-4z} + C_2 e^{5z}$$

$$\therefore \quad y = C_1 \frac{1}{x^4} + C_2 x^5$$

Ex. 3.6.4 : *Solve* $(x^2 D^2 - 7xD + 12)y = x^2$.

GTU : Winter-22, Maths-3, Marks 7
Winter-22, Maths-2, Marks 4

Sol. : Here, $(x^2 D^2 - 7xD + 12)\, y = x^2$ is given

This is Euler's Cauchy equation,

$$\therefore \quad x = e^z \Rightarrow z = \log x, \text{ where, } D = \frac{d}{dx}$$

$$\therefore \quad x \cdot \frac{dy}{dx} = Dy$$

$$\therefore \quad x^2 \frac{d^2y}{dx^2} = D(D-1)y$$

$\therefore$ Given equation,

$$(D(D-1) - 7D + 12)y = e^{2z} \quad \ldots(1)$$

A.E. $D(D-1) - 7D + 12 = 0$

$$\therefore \quad D^2 - 8D + 12 = 0$$

$$\therefore \quad (D-6)(D-2) = 0$$

$$\therefore \quad D = 6 \text{ and } D = 2$$

$$y_C = C_1' e^{6z} + C_2 e^{2z}$$

Let us consider trial solution,

$$y_P = Ce^{2z}$$

Here, e^{2z} appears once in y_C.

$\therefore$ We multiply the corresponding term by z.

$$\therefore \quad y_P = Cz e^{2z}$$

$$y_P' = Ce^{2z} + 2Ce^{2z} z$$

$$y_P'' = 2Ce^{2z} + 2Ce^{2z} + 4Cze^{2z}$$

$$\therefore \quad y_P'' = 4Ce^{2z} + 4Cze^{2z}$$

Now, Put y_P, y_P' and y_P'' in given equation (1)

$$\therefore \quad (D^2 - 8D + 12)\, y = e^{2z}$$

$$\therefore \quad y_P'' - 8y_P' + 12y_P = e^{2z}$$

$$\therefore \quad 4Ce^{2z} + 4Cze^{2z} - 8Ce^{2z} - 16Cze^{2z} + 12Cze^{2z} = e^{2z}$$

$$-4Ce^{2z} = e^{2z}$$

$$\Rightarrow \quad -4C = 1 \Rightarrow C = -\frac{1}{4}$$

$$\therefore \quad y_P = -\frac{1}{4} ze^{2z}$$

$$y = y_C + y_P = C_1 e^{6z} + C_2 e^{2z} - \frac{1}{4} ze^{2z}$$

$$\therefore \quad y = C_1 x^6 + C_2 x^2 - \frac{1}{4} \log x \cdot x^2$$

Ex. 3.6.5 : *Solve* $x^2 y'' - xy' + y = \sin(\log x)$.

GTU : Summer-17, Maths-3, Marks 7

Sol. : Here, $x^2 y'' - xy' + y = \sin(\log x)$ is given

This is Euler's Cauchy equation,

Let us consider $x = e^z \Rightarrow z = \log x$

Now, $\quad x \cdot y' = Dy, \quad D = \frac{d}{dz}$

$$x^2 y'' = D(D-1)y$$

$\therefore$ We write given equation,

$$(D(D-1) - D + 1)y = \sin z$$

$$\therefore \quad (D^2 - 2D + 1)\, y = \sin z$$

$\therefore$ A.E. $D^2 - 2D + 1 = 0$

$$\therefore \quad (D-1)^2 = 0$$

$$D = 1, 1$$

$$\text{C.F.} = (C_1 + C_2 z)\, e^z$$

Here, we use U.C. for finding P.I.

Let us consider trial solutions y_P

$$\therefore \quad y_P = C_1 \cos z + C_2 \sin z$$

$$y_P' = -C_1 \sin z + C_2 \cos z$$

$$y''_P = -C_1 \cos z - C_2 \sin z$$

Put y_P, y'_P and y''_P in given equation,

$$\therefore\ y''_P - 2y'_P + y_P = \sin z$$

$$-C_1 \cos z - C_2 \sin z + 2C_1 \sin z - 2C_2 \cos z + C_1 \cos z + C_2 \sin z = \sin z$$

$$2C_1 \sin z - 2C_2 \cos z = \sin z$$

Comparing coefficient both sides,

$$\therefore\quad 2C_1 = 1 \Rightarrow C_1 = \frac{1}{2}$$

$$-2C_2 = 0 \Rightarrow C_2 = 0$$

$$\therefore\quad y_P = \frac{1}{2}\cos z$$

$$\therefore\quad y = y_C + y_P = (C_1 + C_2 z)\,e^z + \frac{1}{2}\cos z$$

$$\therefore\quad \mathbf{y = (C_1 + C_2 (\log x))\, x + \frac{1}{2}\cos(\log x)}$$

Ex. 3.6.6 : *Solve* $(x^2 D^2 - 3xD + 4)y = x^2$ **GTU : Summer-12, Maths-3, Marks 7**

Sol. : Here, $(x^2D^2 - 3xD + 4)\, y = x^2$

Let, $x = e^z \Rightarrow z = \log x$

and $x \cdot \frac{dy}{dx} = Dy, \quad \frac{x^2 d^2 y}{dx^2} = D(D-1)y$

where, $D = \frac{d}{dz}$

$\therefore$ Given equation can be written as,

$$(D(D-1) - 3D + 4)y = e^{2z}$$

$$\therefore\quad (D^2 - 4D + 4)y = e^{2z}$$

Now, A.E. $D^2 - 4D + 4 = 0$

$$(D-2)^2 = 0$$

$$\therefore\quad \text{C.F.} = (C_1 + C_2 z)\,e^{2z}$$

Here, we use U.C. method to find P.I.

$\therefore$ Let us consider trial solution y_P

$$y_P = Ce^{2z}$$

Here, e^{2z} term appears twice in C.F.

$\therefore$ We multiply z^2 in y_P

$$\therefore\quad y_P = Cz^2 e^{2z}$$

$$y'_P = 2Cz e^{2z} + 2Cz^2 e^{2z}$$

$$\therefore \quad y''_P = 2Ce^{2z} + 4Cz e^{2z} + 4Cz e^{2z} + 4Cz^2 e^{2z}$$

$$\therefore \quad y''_P = 2Ce^{2z} + 8Cze^{2z} + 4Cz^2 e^{2z}$$

$\therefore$ Put y_P, y'_P and y''_P in given equation,

$$y''_P - 4y'_P + 4y_P = e^{2z}$$

$$\therefore \quad 2Ce^{2z} + 8Cz e^{2z} + 4Cz^2 e^{2z} - 8Cz e^{2z} - 8Cz^2 e^{2z} + 4Cz^2 e^{2z} = e^{2z}$$

$$2Ce^{2z} = e^{2z}$$

$$\therefore \quad 2C = 1 \Rightarrow C = \frac{1}{2}$$

$$\Rightarrow \quad y_P = \frac{1}{2} z^2 e^{2z}$$

$$\therefore \quad y = y_C + y_P = (C_1 + C_2 z)e^{2z} + \frac{1}{2} z^2 e^{2z}$$

$$\therefore \quad \mathbf{y = (C_1 + C_2 \log\ x)\, x^2 + \frac{1}{2} x^2 (\log\ x)^2}$$

Ex. 3.6.7 : *Solve* $x^2 y'' - 2xy' + y = \cos(\log x)$

GTU : Summer-20, Maths-2, Marks 4; Summer-23, Maths-2, Marks 3.5

Sol. : Here, $x^2 y'' - 2xy' + y = \cos(\log x)$ is given

This is Cauchy's Euler equation,

$$x = e^z \Rightarrow z = \log x$$

$$xy' = Dy \text{ and } x^2 y'' = D(D-1)y$$

where, $$D = \frac{d}{dz}$$

$$\therefore \quad (D(D-1) - 2D + 1)y = \cos z$$

$$\therefore \quad (D^2 - 3D + 1)\,y = \cos z$$

A.E. $$D^2 - 3D + 1 = 0$$

$a = 1, \quad b = -3, \quad c = 1$

$$\Delta = b^2 - 4ac = 9 - 4 = 5$$

$$D_1, D_2 = \frac{-b + \sqrt{\Delta}}{2a} = \frac{3 \pm \sqrt{5}}{2}$$

$$\therefore \quad \text{C.F.} = y_P = C_1 e^{\frac{3+\sqrt{5}}{2} z} + C_2 e^{\frac{3-\sqrt{5}}{2} z}$$

For finding y_P we use U.C. method,

$$y_P = A \cos z + B \sin z$$

$$y'_P = -A \sin z + B \cos z$$

$$y_P'' = -A\cos z - B\sin z$$

Put, y_P, y_P', y_P'' in given equation.

$$y_P'' - 3y_P' + y_P = \cos z$$

$$-A\cos z - B\sin z + 3A\sin z - 3B\cos z + A\cos z + B\sin z = \cos z$$

$$3A\sin z - 3B\cos z = \cos z$$

Comparing co-efficient both sides

$$3A = 0 \Rightarrow A = 0$$

$$-3B = 1 \Rightarrow B = -\frac{1}{3}$$

$$\therefore \quad y_P = -\frac{1}{3}\sin z$$

$$y = y_C + y_P$$

$$y = C_1 e^{\frac{3+\sqrt{5}}{2}z} + C_2 e^{\frac{3-\sqrt{5}}{2}z} - \frac{1}{3}\sin z$$

$$\mathbf{y = C_1 x^{\frac{3+\sqrt{5}}{2}} + C_2 x^{\frac{3-\sqrt{5}}{2}} - \frac{1}{3}\sin(\log x)}$$

Ex. 3.6.8 : *Solve* $x^3y''' + x^2y'' = x^3$. **GTU : Winter-23, Maths-2, Marks 4**

Sol. : This is Euler Cauchy's equation,

$$x^3y''' + x^2y'' = x^3$$

Let $\quad x = e^z \Rightarrow z = \log x$

and $\quad xy' = Dy$

$$x^2y'' = D(D-1)y$$

$$x^3y''' = D(D-1)(D-2)y$$

where, $\quad D = \frac{d}{dz}$

$$(D(D-1)(D-2) + D(D-1)y = e^{3z}$$

$$(D^3 - 2D^2 + D)y = e^{3z}$$

A.E. $\quad D(D-1)(D-2+1) = 0$

$$D(D-1)(D-1) = 0$$

$D = 0, 1, 1$

$$y_C = \text{C.F.} = C_1 e^z (C_2 + C_3 z)$$

Now, let us consider trial solution y_P

$$\therefore \quad y_P = Ce^{3z}$$

$$y'_P = 3Ce^{3z},\ y''_P = 9Ce^{3z} \text{ and } y'''_P = 27Ce^{3z}$$

Put y''_P, y'''_P in $(D^3 - 2D^2 + D)y = e^{3z}$

$$\therefore \quad y'''_P - 2y''_P + y'_P = e^{3z}$$

$$27Ce^{3z} - 18Ce^{3z} + 3Ce^{3z} = e^{3z}$$

$$30Ce^{3z} - 18Ce^{3z} = e^{3z}$$

$$\therefore \quad 12C = 1 \Rightarrow C = \frac{1}{12}$$

$$\therefore \quad y_P = \frac{1}{2}e^{3z}$$

$$\therefore \quad y = y_C + y_P$$

$$y = C_1 + e^z(C_2 + C_3 z) + \frac{1}{12}e^{3z}$$

$$\therefore \quad \mathbf{y = C_1 + x(C_2 + C_3 \log x) + \frac{1}{12}x^3}$$

Ex. 3.6.9 : *Solve* $(x^2D^2 + xD + 1)y = \log x \cdot \sin(\log x)$ **GTU : Summer-18, Maths-3, Marks 4 Winter-23, Maths-3, Marks 7**

Sol. : Here, $(x^2D^2 + xD + 1)\ y = \log x \cdot \sin(\log x)$ is given

This equation is Cauchy's linear equations,

Let, $x = e^z \Rightarrow z = \log x$

$$\therefore \quad x\frac{dy}{dx} = Dy$$

$$x^2\frac{d^2y}{dx^2} = D(D-1)y, \text{ where } D = \frac{d}{dz}$$

We write given equation in the form

$$(D(D-1) + D + 1)y = z \sin z$$

For, C.F. $\quad D(D-1) + D + 1 = 0$

$$D^2 - D + D + 1 = 0$$

$$\therefore \quad D^2 + 1 = 0$$

$$\therefore \quad D^2 = -1$$

$$\therefore \quad D = \pm i$$

$$\text{C.F.} = C_1 \cos z + C_2 \sin z$$

Now, we use variation of parameter for finding P.I.

$$\therefore \quad y_1 = \cos z \text{ and } y_2 = \sin z$$

$$W = \begin{vmatrix} \cos z & \sin z \\ -\sin z & \cos z \end{vmatrix} = \cos^2 z + \sin^2 z = 1$$

$$W_1 = -y_2 = -\sin z$$

$$W_2 = y_1 = \cos z$$

Now, $$\text{P.I.} = y_1 \int \frac{W_1}{W} r(x)\,dx + y_2 \int \frac{W_2}{W} r(x)\,dx$$

where, $$r(x) = z \sin z$$

$\therefore$ $$\text{P.I.} = \cos z \int \frac{-\sin z}{1} \cdot z \sin z\,dz + \sin z \int \frac{\cos z}{1} \cdot z \sin z\,dz$$

$$= -\cos z \int z \cdot \sin 2z\,dz + \frac{\sin z}{2} \int \sin 2z \cdot z\,dz$$

$$= -\cos z \int z \cdot \left(\frac{1-\cos 2z}{2}\right) dz + \frac{\sin z}{2} \int z \sin 2z\,dz$$

$$= -\cos z \left[\int \frac{z}{2}\,dz - \frac{1}{2}\int z \cdot \cos 2z\right] + \frac{\sin z}{2} \int z \cdot \sin 2z\,dz$$

Now $$\int z \cos 2z dz = z\frac{\sin 2z}{2} - \left(\frac{-\cos 2z}{4}\right)$$

$$= \frac{z \sin 2z}{2} + \frac{\cos 2z}{4}$$

$$\int z \sin 2z\,dz = z\frac{\cos 2z}{2} - \left(\frac{\sin 2z}{4}\right)$$

$\therefore$ $$\text{P.I.} = -\cos z\left[\frac{z^2}{4} - \frac{1}{4} z \sin 2z - \frac{1}{8}\cos 2z\right] + \frac{\sin z}{2}\left[\frac{z \cos 2z}{2} - \frac{\sin 2z}{4}\right]$$

$$= -\frac{1}{4} z^2 \cos z + \frac{1}{4} z \cdot \cos z \cdot \sin 2z + \frac{1}{8} \cos z \cdot \cos 2z + \frac{1}{4} z \cdot \sin z \cdot \cos 2z - \frac{1}{8} \sin z \cdot \sin 2z$$

$\therefore$ $$y = \text{C.F.} + \text{P.I.}$$

Ex. 3.6.10 : *Solve the differential equation* $x^2 y'' - 2xy' + 2y = x^3 \cos x$.

GTU : Summer-19, Maths-2, Marks 7

Sol. : Here, $x^2 y'' - 2xy' + 2y = x^3 \cos x$ is given

Let, $x = e^z \Rightarrow z = \log x$

$$x\frac{dy}{dx} = Dy$$

$$x^2 \frac{d^2y}{dx^2} = D(D-1)y \quad D = \frac{d}{dz}$$

$$(D(D-1) - 2D + 2)y = x^2 \cos x = e^{3z} \cos(e^z)$$

$$(D^2 - 3D + 2)y = e^{3z} \cos(e^z)$$

A.E.
$$D^2 - 3D + 2 = 0$$
$$(D-2)(D-1) = 0$$
$$D = 2, \quad D = 1$$
$\therefore$
$$\text{C.F.} = y_C = C_1 e^{2z} + C_2 e^{z}$$
Compare with
$$\text{C.F.} = C_1 y_1 + C_2 y_2$$
$$y_1 = e^{2z} \quad \text{and} \quad y_2 = e^{z}$$
$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{2z} & e^{z} \\ 2e^{2z} & e^{z} \end{vmatrix} = e^{3z} - 2e^{2z} = -e^{z} \neq 0$$
$$W_1 = -y_2 = -e^{z}$$
$$W_2 = y_1 = e^{2z}$$
$$y_P = y_1 \int \frac{W_1}{W} r(x)\, dz + y_2 \int \frac{W_2}{W} r(z)\, dz$$
$$= e^{2z} \int \frac{-e^{z}}{-e^{z}} e^{3z} \cos(e^{z})\, dz + e^{z} \int \frac{e^{2z}}{-e^{z}} e^{3z} \cos(e^{z})\, dz$$
$$= e^{2z} \int e^{3z} \cos(e^{z})\, dz$$
$$= e^{z} \int e^{4z} \cos(e^{z})\, dz$$

Now, $\int e^{3z} \cos(e^{z})\, dz$

Take
$$e^{z} = t \Rightarrow e^{z} dz = dt$$
$$\int (e^{z})^2 \cos(e^{z}) e^{z} dz = \int \underset{u}{\underset{\uparrow}{t^2}} \underset{v}{\underset{\uparrow}{\cos}}\ t\ dt = t^2(-\sin t) - (2t)(-\cos t) + 2(+\sin t)$$
$$= \int e^{4z} \cos(e^{z})\, dz = \int (e^{z})^3 \cos(e^{z}) e^{z} dz$$
Let,
$$e^{z} = t \Rightarrow e^{z} dz = dt$$
$$\int t^3 \cos t\, dt = (t^3)(-\sin t) - (3t^2)(-\cos t) + (6t)(\sin t) - 6\cos t$$
$$= -t^3 \sin t + 3t^2 \cos t + 6t \sin t - 6\cos t$$
$\therefore$
$$y_P = e^{2z}(-e^{2z} \sin(e^{z}) + 2e^{z} \cos(e^{z}) + 2\sin(e^{z}))$$
$$- e^{z}\ (-e^{3z} \sin Ce^{z}) + 3e^{2z} \cos(e^{z}) + 6e^{z} \sin(e^{z}) - 6\cos(e^{z}))$$
$$= -e^{4z} \sin(e^{z}) + 2e^{3z} \cos(e^{z}) + 2e^{2z} \sin(e^{z}) + e^{4z} \sin(e^{z})$$
$$-3e^{3z} \cos(e^{z}) - 6e^{2z} \sin(e^{z}) + 6e^{z} \cos(e^{z})$$
$$= -e^{3z} \cos(e^{z}) - 4e^{2z} \sin(e^{z}) + 6e^{z} \cos(e^{z})$$
$$y = y_C + y_P$$
$$= C_1 e^{2z} + C_2 e^{z} - e^{3z} \cos(e^{z}) - 4e^{2z} \sin(e^{z}) + 6e^{z} \cos(e^{z})$$
$$\mathbf{y = C_1 x^2 + C_2 x - x^3 \cos x - 4x^2 \sin x + 6x \cos x}$$

Ex. 3.6.11 : *Solve the differential equation :*
$x^2 \dfrac{d^2y}{dx^2} - 4x\dfrac{dy}{dx} + 6y = x^2 \log x$ *using method of variation of parameters.*

Sol. : This is a Cauchy's linear equation with variable coefficients.

Putting $x = e^t \quad \therefore \quad \log x = t$

$$\Rightarrow \qquad x\frac{dy}{dx} = Dy, \quad x^2\frac{d^2y}{dx^2} = D(D-1)y$$

$\therefore$ Given differential equation may be rewritten as

$$(D(D-1) - 4D + 6)y = te^{2t}$$

$$\Rightarrow \qquad (D^2 - 5D + 6)y = te^{2t}$$

Auxiliary equation is : $(m-2)(m-3) = 0$

$$\Rightarrow \qquad m = 2, 3$$

$$\text{C.F.} = C_1e^{2t} + C_2e^{3t} = C_1y_1 + C_2y_2$$

$\therefore \quad y_1 = e^{2t}$ and $y_2 = e^{3t}$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{2t} & e^{3t} \\ 2e^{2t} & 3e^{3t} \end{vmatrix} = e^{5t}$$

$$u_1 = -\int \frac{y_2F(t)}{W}dt = -\int \frac{e^{3t}te^{2t}}{e^{5t}}dt = -\int t\,dt = -\frac{t^2}{2}$$

$$u_2 = \int \frac{y_1F(t)}{W}dt = \int \frac{e^{2t}te^{2t}}{e^{5t}}dt = -\int te^{-t}\,dt = [(t)(-e^{-t}) - (1)(e^{-t})] = -te^{-t} - e^{-t}$$

$$\therefore \qquad \text{P.I.} = u_1y_1 + u_2y_2 = -\frac{t^2}{2}e^{2t} - (te^{-t} + e^{-t})e^{3t}$$

$$= -\frac{t^2}{2}e^{2t} - te^{2t} - e^{2t} = -e^{2t}\left(\frac{t^2}{2} + t + 1\right)$$

Complete solution is : $\quad y = \text{C.F.} + \text{P.I.}$

$$\Rightarrow \qquad y = C_1e^{2t} + C_2e^{3t} - e^{2t}\left(\frac{t^2}{2} + t + 1\right)$$

or

$$y = C_1x^2 + C_2x^3 - x^2\left(\frac{(\log x)^2}{2} + \log x + 1\right)$$

$$\Rightarrow \qquad y = C_3x^2 + C_2x^3 - \frac{x^2}{2}(\log x)^2 - x^2\log x, \quad C_3 = C_1 - 1$$

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